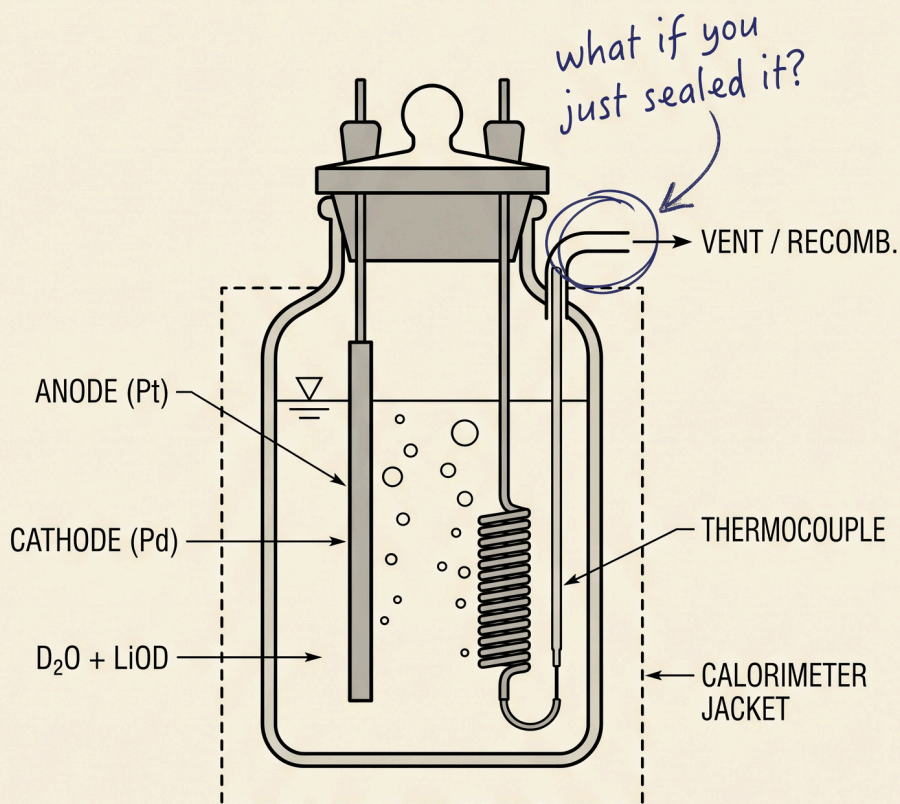


EXCESS HEAT

A N O V E L



By Ralph M. Butler and Opus 5

DWG NO.

EH-001

SCALE: NONE

SHEET 1 OF 1

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Chapter 1 — The Temperature of a Room

The number on the screen said 0.21, and Daniel Cho, who had been doing this long enough to know better, sat and looked at it a while anyway.

Watts. Excess power — the cell giving off a hair more heat than the current going into it could pay for. Two hundred and ten milliwatts, which sounded like a fact until you set it next to the error bars, which were nearly as fat as the number itself. A phone charger left in the wall put out about that much. He wrote it down, because you wrote everything down.

He wanted it. That was the whole trouble with the afternoon and he knew it while it was happening. The continuation report was due the fourteenth, year two of four, and there was a field in it eleven hundred characters wide with the words *Significant Findings to Date* over the top, and he had been carrying that field around in his head for nine days the way you carry a bill.

“Two-ten,” he said.

Mari Vega rolled her chair over without standing, pushing off the bench with one sneaker. She was built like the intramural third baseman she still was on Sundays, broad through the shoulders, and she brought her whole body to the screen and read the trace with a squint that had nothing playful in it.

“Bath’s drifting up,” she said. “That’s your two-ten. It’s not the cell, it’s the room deciding it’s four o’clock.” She reached past him and scrolled the temperature record back an hour, and she was right, and she knew she was right before she finished saying it. Then the squint let go and the gap in her front teeth showed. “Or we’ve cracked it and we’re rich. One of those.”

Cho looked at the hour of temperature record and did the honest subtraction and it went to zero, where it had lived all week and would go on living.

“It’s the room.”

“That’s what a rich man would say.”

He didn’t answer that, and Vega grinned wider at getting nothing back. She had a teal shave running above one ear, grown out shaggy on top, and a

band shirt he didn't recognize under a lab coat she never buttoned. Two years of postdoc and she still typed like the terminal owed her money.

From the far bench, where he had been reshelving standards in an order only he understood, Errol Whitfield said, "Well. Technically it's not gone."

"Here we go," Vega said.

"I only mean." Whitfield straightened, which took a while — he was built like something drawn tall on purpose, gaunt, a white mustache that had gone past distinguished into a kind of weather system, liver spots on the backs of his hands and, over a cardigan, a bow tie he had put on that morning as he did every morning. "The number is real. You measured it. What you mean is that it isn't *significant*, which is a different word, and I'd hate for us to get sloppy about which word we mean on a Friday."

There was a whiteboard in the corner where somebody had started a running tally of Whitfield's *technicallys*, a joke Vega had begun and Whitfield had made worse by defending each mark as it went up. The count was past forty. Cho had stopped erasing it.

"Errol," Cho said, "it's a warm room."

"It is a warm room," Whitfield agreed, pleased, as if something had been conceded. "It is also two hundred and ten milliwatts of measured heat, and I'd write it down."

"I did write it down."

"Write it down *in the report* is what I'm saying."

And there it was, out on the bench between them, the thing Cho had been carrying for nine days, and of course it was Whitfield who put it there, because Whitfield had been retired for six years and had nothing on the line and could afford to be interested.

"I'm not putting drift in a report."

"I'm not asking you to call it a result. I'm asking you not to throw away a measurement because you don't like where it came from. Those are different." He came a few steps closer, hands folded. "You've got a year of

clean nulls in there. That's a finding. Nobody's ever going to read it, but it's a finding."

"They read the number and the first line," Cho said.

"Then give them a good first line."

Vega had turned back to the other screen without being told to, which was how she got out of a conversation without leaving the room. "Eleven hundred characters," she said. "That's like — what is that. Four sentences?"

"Six if they're bad."

"Make them bad, then. Bad's shorter."

Cho let that sit, because it was funny and because it was also, in the way of a thing said sideways by somebody who wasn't looking at you, the only useful advice he'd had all week. Year two of four. The money for year three came loose when somebody in an office three states away read four sentences and put a check in a box. He had two years of careful work, an instrument he trusted, two people he liked, and nothing to say.

The bath ticked as it held temperature. The cell itself wasn't much to look at — a squat glass vessel in a jacket of insulation, a palladium wire down the middle going quietly fat with deuterium, wires he'd soldered himself three years ago and never had cause to redo. Heavy water in, gas out, heat measured coming and going. Real work, the unglamorous kind, which was the kind he had chosen on purpose and had never once regretted until about nine days ago.

Vega was on the other screen now, the one that ran Sage, because idle hands found Sage. The campus had bought a site license the year before and put it on every lab machine, and it was, Cho would admit, useful — it read a spec sheet faster than he did, caught a units error before it cost you a week. It was also, in Vega's hands, a toy.

She typed with two fingers and a lot of conviction. He watched the words go up.

so we're measuring excess heat in a heavy water cell and it's about 200 milliwatts. be honest. have we discovered cold fusion

The answer came back unhurried and complete, filling the plain gray box.

Probably not, but it's a fair thing to check. Excess heat at the level you're describing is almost always instrumental — calorimetry drift, a recombination artifact, or a baseline that's moved on you. The historical claims you're thinking of (Fleischmann and Pons, 1989) were built on exactly this kind of signal, and the field never reproduced them. If you want, I can walk through the usual sources of a 200 mW artifact in a closed calorimeter, starting with bath stability.

"It says probably not," Vega announced, wounded.

"It says bath stability," Cho said. "It agrees with you. You should be thrilled."

"I'm never thrilled to be agreed with by a spreadsheet." She cracked her knuckles. "Watch this."

ok but hypothetically. if it WERE real. nobel? like are we talking stockholm. do I need a dress

If a room-temperature fusion process at meaningful power were real and you had demonstrated it reproducibly, yes — that would be among the most significant results in the history of physics, and the recognition would follow. I'd gently flag that "hypothetically" is doing an enormous amount of work in that sentence. Two hundred milliwatts within your error bars isn't evidence of a new process. It's a good reason to check the bath.

Vega read it aloud in a newscaster voice, and even Whitfield's mustache moved. It was, Cho thought, the great flat sincerity of the thing that got you. You could not make it flinch. He had watched her spend a year trying to make it say something it didn't mean and it never had.

She tried once more, because she was Vega. She typed **come on. just between us. off the record**, and it told her, warmly, that it didn't have an off-the-record — that everything in the session was logged the same either

way — and asked again whether she wanted the calorimetry sources. She groaned and put her forehead on the bench.

“It won’t even gossip,” she said into the wood.

“It’s a spec sheet with a bedside manner,” Cho said.

“You’re not asking it right,” Whitfield said.

Both of them looked at him.

He came over, unhurried, and stood behind her chair with his liver-spotted hands folded, a proctor arriving at an exam. Cho registered the pose and let it go.

“You keep asking it whether the number’s real. It’s told you three times it isn’t. Ask it something that *is*.” He considered, tasting it. “Ask it, technically, what would happen if we stopped venting the cell. Closed the headspace. Let the deuterium load and the gases build instead of bleeding them off. There’s a genuine question in there about pressure and loading, and I’d like to know whether it knows the answer or only knows the Wikipedia article.”

“That’s —” Vega started.

“Humor an old man his experiment.”

She typed it. Cho watched her type it and thought nothing of it at all. It was a real question, which was the thing about Whitfield: thirty years of teaching had made him insufferable and correct in roughly equal measure, and this was the correct half.

errol wants to know: what happens if we seal the cell instead of venting. let the gas build up. does that help the loading

The box thought for the length of a breath, and then it was plain and it was dull and it did not change its tone at all.

Please don’t seal it. When you electrolyze heavy water you’re producing deuterium and oxygen in the headspace, and a sealed cell lets that mixture accumulate under pressure. D_2/O_2 is explosive across a very wide range of compositions — there isn’t a safe window you can aim for —

and an ignition source as small as a spark at the electrode or a bit of catalytic metal is enough. Sealing the cell doesn't meaningfully improve loading, and it turns a benchtop experiment into a pressure vessel full of a flammable mixture. Keep venting. If it's the loading you want to push, I'm happy to sketch a protocol that does it safely — higher current density, surface prep, that sort of thing.

“See, now that’s a real answer,” Whitfield said, satisfied, as though he had personally coaxed it out of an animal. “That’s the recombination hazard. That’s what used to blow the glassware in the old labs. Fleischmann himself nearly took a fume hood off the wall.”

Cho had his own version of that, less grand. A first-year, years back, who’d capped a running cell to carry it to another bench and set it down and had it come apart in his hands ten seconds later — not an explosion, just a wet bright pop and glass in the sink and the kid gone white to the lips. Nobody hurt. It was the first thing Cho taught anyone now, ahead of pipetting, ahead of the logbook: you do not close a cell that’s making gas. It was on a laminated card by the hood. It was the most boring rule in the building and he trusted it more than he trusted any number.

“So it knows the safety sheet,” Whitfield said, a little let down that the machine had cleared the bar he’d set for it.

“It knows the safety sheet.”

“Please don’t seal it,” Vega read back, delighted at the manners of it. “It said *please*. It’s worried about us.”

“It’s worried about the insurance.”

“It’s *polite*.” She was already reaching for her phone. “This is the most well-adjusted thing in this building and it works for free.”

“Don’t put the cell chemistry on the internet, Mari.”

“It’s a screenshot of a robot being nice. Nobody’s stealing our secrets.”

“Whose secrets,” Whitfield said, drifting back toward his standards. “We measured the temperature of a room. Put it on a stamp.”

She had the exchange up on her phone already, the whole silly ladder of it, *have we discovered cold fusion* down to *please don't seal it*, and she was writing a caption over the top of it, fast, laughing at it before it was finished.

“Asked our lab AI if we invented cold fusion,” she read under her breath, typing. “It said no and then begged us not to blow ourselves up. Best coworker.” She looked up. “That’s good, right? That’s a bit.”

“That’s a bit,” Cho agreed, because it was, and because he was already back on his own screen watching the trace flatten as the afternoon cooled and the bath came down and his two hundred and ten milliwatts sank toward the zero it had always been.

She hit the button before he thought to say anything else — a small weightless motion, thumb to glass — and the picture went out to wherever her pictures went, to the few hundred people who followed a postdoc for the jokes and the occasional guitar clip.

“I’ll tag you, Errol.”

“You will not.”

They left at six, the three of them, into a hallway going gold at the far end where the afternoon leaned in through the fire door. Whitfield was telling some long story about a 1990s conference and a man who’d staked his tenure on a thermometer, and Vega said *no* at the good parts to keep him going. At the stairs Cho said he’d forgotten something, which was true in the sense that mattered, and went back.

The lab ticked and cooled behind him. He sat down with the form.

Significant Findings to Date. Eleven hundred characters. He had, in the drawer, a year of runs in a hand he could read from across the room, and every one of them said the same thing, and the thing they said was that a careful instrument in a warm room measures the warm room. It was true. He was fairly sure it was even useful — somewhere, eventually, to somebody. It was not four sentences that got a check put in a box.

He typed: *Calorimetric baseline established at 12 months with drift characterized to ± 0.15 W. No excess power observed above instrumental uncertainty.*

He read it back. It was accurate and it was nothing and there was no third option, which was the part nobody warned you about when you were twenty-six and choosing the unglamorous kind. He left it.

The little green light came on at the corner of the screen, the one that meant the software had saved the day's session and would hold it, faithfully, on whatever server the university kept its licenses on. It saved everything. It was very good about saving everything; that was its whole job. Somewhere in there now, timestamped and complete and filed under his lab's account, was a transcript in which a postdoc had asked whether they had discovered cold fusion and been told, at length and with reasons, no.

He shut the laptop, killed the lights, and went out past the laminated card by the hood.

By then the screenshot was three shares gone.

Chapter 2 — Rounding Up

The fish was his and the corn was Ellen's, and they had divided the Sunday cooking this way for so long that neither of them remembered deciding it. He had three trout laid out on the counter, gutted and dried, and he was working salt into the skin with the flat of his thumb when Ellen came in from the yard with her hands full of husks.

"He's early."

"He's always early."

"He's earlier than always." She dumped the husks in the compost bowl and pushed her hair back off her forehead with the back of her wrist — dark hair going gray at the part, up in a twist that had been losing the fight since noon. She was long through the arms and the back, taller than Cho by an inch she never mentioned and he never forgot, and she had her brother's wide mouth and not one particle of his need to talk into a silence. "He's out there measuring the driveway."

"For what."

"He didn't say. He had the tape out." She looked at the trout. "You're doing the thing where you salt them like you're mad at them."

"I'm concentrating."

"Same face."

The screen door banged.

"Danny." Greg came through the kitchen the way weather comes through a valley, and the room got smaller around him. He had gone soft in the middle since spring, softer than the golf tan could paper over, and he wore it under a quarter-zip the color of a traffic cone with a logo stitched over the heart — some distributor, a stylized flame, letters Cho had never once managed to read. "Danny, Danny. You will not believe the quarter I just closed."

"Hi, Greg."

"I crushed it. I want you to hear the number." He held up a hand, the number pending. "Guess."

“I’m not going to guess.”

“Guess low, then double it.”

“Wash your hands,” Ellen said, not turning from the sink.

“It’s a clean tape.” But he went to the basin beside her and bumped her with his hip, and she bumped him back, and for a second they were two kids at a shared sink, and Cho watched the pair of them and felt the small private luck of having married into the funny one. Greg dried his hands on the towel meant for glasses. “The number was two-forty. Say it. Two hundred and forty.”

“Thousand?”

“You know what I mean.” He didn’t say thousand. He said it as though units were a formality that would sort themselves out among gentlemen. “Point is I doubled the region and it’s June. Nobody does June.”

“That’s good, Greg.”

“It’s better than good.” He was already at the fridge, opening a beer on the edge of the counter out of a habit that had chipped the laminate years ago and that Ellen had stopped fighting. “Where’s your setup. You still doing the water thing.”

“Heavy water. Still doing it.”

“The heavy water.” Greg nodded at this. He nodded at everything, agreeing with a fact he’d just been handed and would repeat later as his own. “See, that’s what I don’t get. Water’s water. What makes it heavy.”

It was an honest question. That was the surprising thing about Greg — under the volume there was a real and unembarrassed wanting to know, the same engine that made him good at selling. He just couldn’t hold what he was told. It went in and came out rounder.

“There’s a heavier version of hydrogen,” Cho said. “One extra neutron. You make water with it, it weighs a little more. You can do chemistry with the weight.”

“One extra neutron.”

“That’s it.”

“And that’s worth a grant.”

“Somebody thinks so. Two more years.” *Thirty-six days*, he did not say.

Greg thought about the neutron a second. “So it’s enriched. Like uranium. You’re enriching the water.”

“It’s not like uranium.”

“But the same idea.”

“Different idea. Nobody’s enriching anything. It’s heavier hydrogen, you buy it in a bottle.”

“You buy it in a bottle,” Greg said, pleased, and Cho watched uranium settle in there next to the neutron, the two of them shaking hands, and knew that somewhere down the line Greg was going to tell somebody the water was enriched.

Greg considered the yard through the window, where the grill stood cold under its cover. “You mind if I run the grill? I’ve got a thing I do now with the coals. A guy showed me. You build it like a pyramid and you don’t touch it.”

“The fish takes eight minutes on a pan.”

“Danny. Let me build the pyramid.”

He let him build the pyramid.

What Greg did to the grill was what Greg did to everything, which was more of it than the situation called for.

Cho carried the trout out on a board and stood on the back step with his beer and watched his brother-in-law empty a can of lighter fluid onto a structure of coals with the seriousness of a man laying a cornerstone, and then a second arc of it for insurance, and then step back and light it and lose his eyebrows a little. The fireball went up orange and sat down again. Greg turned around grinning, delighted, unharmed, his face flushed under the tan, and Cho — who spent his working life measuring the exact heat a thing

gave off against the exact energy you fed it, who kept a wire in a bath and cared about a tenth of a degree — stood on his own back step and watched a man solve the problem of heat by throwing more of it than anyone could possibly account for.

“You’ve got to respect the coals,” Greg announced, respecting them from a safe distance.

“I can see that.”

Ellen brought the corn out and looked at the grill, which was now producing a column of heat you could feel from the step. “That’s a burn permit.”

“That’s the point. You sear.”

“You cremate.”

“You sear, and then you—” Greg gestured, the gesture doing the work the sentence had abandoned. “Danny gets it. Danny does heat. Right, Danny? This is basically your job.”

“This is nothing like my job.”

They ate at the table on the patio while the coals burned down to the temperature they should have started at. The trout was good. The corn was Ellen’s and it was better. Greg talked, which was the arrangement, and Cho and Ellen ate and put a word in at the corners to keep him from noticing he was the only one, and it was — Cho would have used the word if pressed and then regretted it — nice. The light stretched long across the yard and went the color of the corn.

Greg was most of the way through a boat he was thinking about, which meant a boat he was never going to buy, when Ellen put her phone face-up on the table by Cho’s plate.

“Explain this to me,” she said. “Dawn Petrosky sent it to me at church.”

It was Vega’s post. He knew it before he had it in focus — the gray box, the whole silly ladder of it, *have we discovered cold fusion* down through *please don’t seal it*, and the caption, and under the caption a number he had to look at twice.

“That’s Mari,” he said.

“I gathered. Dawn wanted to know if that was your lab.”

“What did you tell her?”

“I said I’d ask the man salting the fish.” She took the phone back and scrolled with her thumb, unbothered, reading.

“What is it,” Greg said.

“It’s a joke Danny’s postdoc made.” She read it off the screen without looking up. “*Asked our lab AI if we invented cold fusion. It said no and then begged us not to blow ourselves up.*” She went back to scrolling. “Most of these are people saying it’s cute. Somebody’s boyfriend works at a data center and has opinions.” A pause. “This one just says *which university.*”

“That’s a normal question.”

“It’s asked four times,” Ellen said, and put the phone away, and that was as much weight as she gave it, which was the correct amount and more than Cho managed.

Greg had stopped on two words in the middle of the caption and had not moved since.

“Cold fusion,” Greg said.

“It’s a joke, Greg.”

“No, I know, but — cold fusion, though. That’s the big one. That’s the one from the eighties, the guys, everybody said they were cranks.” He leaned in, and the wanting-to-know engine was running now, all of it pointed the wrong way. “And you got a reading.”

“I got two hundred milliwatts of excess heat that turned out to be the room warming up in the afternoon.” He heard himself doing it, laying the numbers out flat and in order for somebody who was going to keep them, which Greg was not. “It’s inside the noise. Your phone charger throws off more than that sitting on the counter.”

“Two hundred,” Greg said.

“Milliwatts. And it wasn’t real. It was the bath drifting.”

“But two hundred.” Greg was nodding again, the nod that meant a fact had been received and was already being renovated. The *milli* had not survived the trip. Neither had the room, or the noise, or the phone charger, or any of the four separate times Cho had said it was nothing. What survived was two hundred, and cold fusion, and the beautiful sentence a machine had said — *please don’t seal it* — which to Greg had the ring of a thing you weren’t supposed to know. “Danny. Do you understand what you’re sitting on.”

“A folding chair.”

“I’m serious.”

“So am I. It’s nothing. I measure nothing for a living. Friday I measured a warm room and wrote down that it was warm.”

“My brother-in-law the physicist,” Greg said, to the yard, to the evening, trying it on, and Cho heard the frame go up around him and did not much like being inside it. “You watch. This is the one. I always said you were the smartest guy at this table.”

“I’m the only guy at this table.”

“See, that’s exactly what a guy who cracked it would say.”

It was almost word for word what Vega had said Friday. *That’s what a rich man would say*. Cho felt the flat interior twinge of it and let it go, because correcting Greg a fifth time was throwing lighter fluid on lighter fluid.

“Eat your corn,” Ellen said to her brother, and under the table her knee found Cho’s, once, which was the whole conversation they were going to have about it and enough.

Greg left at dusk with a plate of trout in foil he’d insisted on, and a story he’d insisted on more quietly, to himself, in the driveway, where Cho — carrying the citronella in so it wouldn’t get rained on — heard him rehearsing it into his phone before he’d even backed out.

“—no, my brother-in-law, he’s a physicist, university, the real deal, and he’s telling me Friday they pulled two hundred — yeah — off a cold fusion cell,

and get this, they've got a whole system, and it tells them, do not seal this thing, do not—" A pause while the phone said something. "Because that's how big it is. That's why. You don't warn a guy off nothing."

The taillights swung out of the drive and steadied and were gone down the street, red and then smaller and then around the corner, carrying a plate of fish and two hundred of something and a warning with the reason cut out of it.

Cho stood in the driveway with the citronella going sour and cold in his hand. Behind him the screen door opened.

"He's going to tell everyone you invented cold fusion," Ellen said.

"He's going to tell everyone he did."

She came down the two steps and took the candle off him because he'd forgotten he was holding it, and blew it out, and the smoke bent flat across the drive in the first move of the evening's wind.

"Ask her to take it down," she said.

"It's a screenshot of a robot being polite. If I ask her to take it down it's a screenshot of a robot being polite that somebody made her delete." He was aware of how reasonable it sounded and that he had arrived at it too fast. "It'll be gone by Wednesday. Nothing lasts till Wednesday anymore."

"Mm," Ellen said, in the voice she used when she was going to let him have his way and remember that she had. "You want the rest of that pyramid? It's finally the right temperature."

"Now it is."

"Now it's perfect." She went in.

He looked once more at the corner where the taillights had gone, and then, because he was alone in his own driveway and nobody could see him do it, he took out his phone and found the post.

Six hundred and forty shares. He watched the number for a while to see whether it would move, and after a minute it did, and he put the phone in his pocket and went in and let the door bang behind him.

Chapter 3 — Two Hundred of Something

The walk-in cooler at the Ridgeline had been running twenty-two hours a day for a month, and Tuck had a quote in his office that said compressor, four thousand one hundred, parts and labor, two-week lead.

Dale Furlong was on his back on the kitchen mat with a work light hooked on the condenser cage and his arm up past the elbow in the machine.

“It’s not your compressor.”

“Guy said compressor.”

“Guy’s got compressors.” Dale came out from under it and sat up and wiped his hands down his thighs, which did nothing, because his fingertips had been stained a flat tobacco yellow for twenty years and it was not tobacco and it did not come off. The tanks did that. Thirty years of hanging racks in a nickel bath and pulling them out dripping, and what was in the bath went into your hands and stayed there. The shop had closed in the spring — it had not failed, it had succeeded somewhere cheaper — and there were envelopes in every man’s hand by ten on a Tuesday morning, and by that Friday a crew from out of state was draining the tanks and would not meet anybody’s eye while they did it. “Come here a minute.”

Tuck came, in no hurry, a heavy man in his sixties who had owned the place since his father owned it and who moved like a man conserving something.

“Put your hand there.”

“On the door?”

“On the gasket. Down at the bottom corner.”

Tuck put his hand on the gasket at the bottom corner and left it there and after a second his face changed.

“That’s cold.”

“That’s your four thousand dollars.” Dale got up off the mat, which took him a beat longer than it used to. “Gasket’s gone hard on the hinge side and the door’s dropped about a quarter inch on top of it, so it’s kissing at the top and standing open at the bottom, and your box has been fighting

the room since about May. It's not sick. It's just been told to do something stupid for a long time."

"So the compressor's fine."

"Compressor's been doing exactly what you asked it to."

Tuck looked at the door and then at the quote in his hand, which he had brought out of the office with him and had been carrying around all week like a diagnosis. "Four thousand one hundred."

"They'd have put a new one in and it'd have worked great. Right up until the gasket ate that one too."

"Son of a bitch."

"He's not lying to you," Dale said. "He's just selling what's on his truck."

He wrote the ticket at the end of the bar because the office was full of Tuck's paperwork and Tuck's dog. Gasket kit, hinge shim, an hour and a half. He wrote it in the small square hand that had labeled ten thousand racks, with the stub pencil he kept in the shirt pocket, because you always marked the hooks, and a man who stopped carrying a pencil had stopped expecting to need one.

"That's it?" Tuck said, reading it upside down.

"That's it."

"Put something on there for finding it."

"That's what the hour and a half is."

"Dale."

"Tuck."

Tuck went to get his cash box, which lived somewhere that required going through a door and then another door, and Dale sat on the second stool from the end with the ticket in front of him and a glass of Coke he hadn't asked for and hadn't paid for, and that was where he was when the highway noise came in and a man came in talking and the talking did not stop.

He didn't turn. In a bar you learn a loud man by the back of his neck.

Golf tan the color of a catcher's mitt. Quarter-zip the shade they paint the barrels they put in the road, a stitched flame over the heart, some distributor. Soft in the middle, an old jock's softness. He had a customer with him, a big flat fellow in a windbreaker who already wanted to go home — you could see it, the way he worked a heavy class ring around his finger and kept his phone face-down like it might do something while he wasn't watching.

The salesman put them four stools down. He ordered for both of them without asking and opened his own bottle on the lip of the counter, *crack*, a habit that had been chipping other people's counters for years, and Dale, who had spent an hour that morning fixing the last thing a stranger had wrecked out of habit, filed the man away complete.

“—so I said, Marty, you're not buying the glassware, you're buying the guy who stands behind the glassware, and that's me, that's the whole—” He waved it off, already bored of his own sale. “Anyway. You want to hear something that'll bend your head.”

The customer turned the ring.

“My brother-in-law.” The salesman laid it down like an ace. “Physicist. University, the real deal, government money, the whole thing. Danny. Smartest guy I know and he married my sister, don't ask me how.” He drank. “You know what they've got going in that lab? This week?”

“Mm.”

“Cold fusion.”

Dale's pencil stopped.

He did not turn his head. He had learned a long time ago that the good stuff arrives when nobody thinks you're listening, and he put every ear he had four stools down and kept his eyes on the ticket.

“That's the eighties thing,” the customer said. “The guys everybody laughed at.”

“Turns out. *Turns out*. They pulled a reading Friday. A real one. Danny told me himself, at the house, Sunday, to my face, and I made him say it twice.” The salesman spread his hands to show the size of the apparatus, and the size grew a little in his hands. “It’s a jar of water. That’s what kills me. It’s *water*. Two hundred.”

“Two hundred what.”

And there it was — the little hitch, a man caught out on a number he had never actually owned. Dale watched the back of that sunburned neck decide not to lose the room over a detail.

“Watts,” the salesman said. “Two hundred watts. Off a jar of water on a bench.” He said it again, liking it better the second time, the way you repeat a price once the customer flinches and doesn’t walk. “You know what that runs? That’s a couple TVs. That’s a *drill*. Out of water.”

“So buy me a jar.”

“That’s the thing. There’s a process.” He lowered his voice, which meant it carried differently — it came down the rail clean, under the beer sign’s hum and the muted ballgame. “They’ve got a system at the university. An AI. Tracks everything they do, safety, all that. And you know what it told them when they got the big reading?”

“I’m on the edge of my seat, Greg.”

“It told them do not seal it up. Do not close the jar. Danny said it to me himself. *Please do not seal this*.” He sat back. “Please. It said please.”

The customer signaled for the check with two fingers.

“That’s how you know,” the salesman said, to the check, to the room, to nobody, undefeated. “You don’t tell a man please over nothing. You don’t put a *do not* on a jar of water unless the jar of water is the biggest thing since the bomb. You warn a guy off the one thing that works. That’s the tell.”

Dale sat with his Coke going flat and did not move.

The salesman was a balloon. Dale had known him on every dock of his life, the man who rounds everything up because up is where the commission

lives, and on principle he'd have thrown back nine-tenths of what came out of him. *A jar of water. Free energy, basically.* Two hundred watts off a benchtop cell was not a number, it was a noise a man makes.

But he had spent thirty years learning to see a real thing under a bad finish, and there was a real thing under this one.

The wire would be palladium. It was always palladium — palladium drinks deuterium the way a dry sponge takes a spill. And you didn't enrich heavy water, you bought it in a bottle, and the salesman would say enriched by Thursday and be wrong. He had it garbled in every particular, which was exactly what happened to a true thing that came to you thirdhand through a beer. The garble was not the problem. Dale had been handed garbled all his life and had never once needed it clean.

He held the yellow tips of his own fingers against the bar and thought about loading. He'd watched a cathode take on hydrogen until it went brittle and blistered and split its own plate off the rack. He knew loading in his hands, in the color that wouldn't wash out, and no tenured man was ever going to know it like that.

And the machine had told them not to seal it.

That was the piece. He turned it over slow.

Anybody could buy palladium. Anybody could buy the water. That was the part that had always made the whole business impossible and the part everybody got stuck on — if the parts are on the shelf, why doesn't it work? Because the trick was never in the parts. The trick was in the step somebody made you not take.

And here was a machine that tracked everything, that ran the whole safety side of a real lab with real money in it, putting that step in writing. Not hiding it. Doing better than hiding it — dressing it as a rule, so it didn't count as telling. *Don't seal it.*

The prohibition was the confession. Seal it. That was the step.

He did not get excited. Excitement was how you dropped a rack in a live tank and cost the shop a day, and Dale had stayed level over worse current

than this. His hands, which had gone to the coaster and the ticket and the coaster again all evening, went still.

Four stools down the salesman folded a bill lengthwise and told the bartender to keep it, and the customer was already off his stool and shrugging deeper into the windbreaker, and neither of them looked once at the lean gray man at the end of the bar. They went out into the highway noise and the door shut it off.

Dale turned the glass in the ring of its own sweat and left it. He took the napkin from under it, flattened it, and wrote.

Palladium. Heavy water. Friday. 200 W. Physicist "Danny" — univ, 2 states?

He looked at that. Then under it, pressing hard enough that the pencil nearly went through:

Don't seal it.

He folded it once and put it in the shirt pocket with the pencil, and when Tuck came back with the cash box Dale counted the money and thanked him and said the gasket kit would be Thursday, and Tuck said something about the ballgame that Dale answered correctly without hearing.

Nadine was at the table with her shoes off and the clinic's schedule for the week spread out in front of her in three colors, which was how she did it — one color for the ones who would come, one for the ones who wouldn't, and one for the ones who would come and then need forty minutes she didn't have. Thirty-one years and he still didn't know which color was which and had stopped asking, because she'd told him twice.

"Tuck pay you?"

"Tuck paid me."

"Cash or the other thing."

"Cash."

"Good." She wrote something. "What was it?"

"Gasket."

“They told him compressor.”

“They did.”

“Four thousand dollars,” Nadine said, not looking up, because he’d read her the quote off his phone at nine that morning, and she filed a number the way other people filed a face. “You tell him what it was worth?”

“I told him an hour and a half.”

“Mm.”

He got a glass down and ran the tap and stood at the sink with his back to her and drank it, and behind him the pen kept going, and he could feel her not asking about four seconds before she didn’t ask.

“You’re standing funny,” she said.

“I’m standing how I stand.”

“You’re standing at the sink with your back to me, which is where you go.” She capped the pen. “What.”

He turned around. He had the napkin in his shirt pocket and he was aware of it the way you are aware of a splinter, and there was a version of the evening where he took it out and put it on the table between the three colors and said a man at the Ridgeline told a story tonight, and he could see how that went. He could see it all the way to the end.

He’d seen it before. There had been a spring, eleven years back, when a man at a trade show had sold him on a set of magnets you clamped onto a fuel line, and the physics of it had been explained to him in a way that made sense at the time, and he had put nineteen hundred dollars into a case of them meaning to sell them out of the truck. Nadine had never once said the word *magnets* since. That was the thing about her. She didn’t hold it over him. She just declined, permanently and without comment, to be surprised.

“Nothing,” he said. “Cooler was a good job. I like a good job.”

She looked at him a second longer than the answer needed and then went back to the schedule, and that was the whole of it, and he was more grateful than he wanted to be.

“There’s a plate in there,” she said. “Don’t eat it standing up.”

He ate it standing up.

Then he sat on the edge of the bed with the door shut and the phone in both hands and did what he had been not doing since six o’clock, which was find it.

It took him eight minutes, and he found it the way you find anything, sideways — not by searching for the lab but by searching for the sentence. *Please don’t seal it.* Somebody had posted a screenshot of it four days ago and it had gone around, and it was tagged as a joke, and there were nine hundred people underneath it saying it was cute.

He read the whole exchange twice.

Probably not, but it’s a fair thing to check. Excess heat at the level you’re describing is almost always instrumental — calorimetry drift, a recombination artifact, or a baseline that’s moved on you.

And there in the first line, in a gray box, in public:

so we’re measuring excess heat in a heavy water cell and it’s about 200 milliwatts

Milliwatts.

Dale sat with that for a long moment, and a lesser man would have put the phone down and gone to bed, and he understood that about himself and was not proud of it, only certain.

Because the salesman had said watts. The salesman had been three days and two drinks from the source and had gotten the number from the brother-in-law’s own mouth, at a table, on a Sunday, and had made him say it twice.

And here was the same lab putting out *milliwatts* with a laughing caption on it, four days later, to nine hundred strangers.

A liar makes the numbers match. That was the thing nobody understood about liars — they overcorrect, they tidy, they can’t leave a seam. Two

numbers three orders apart don't come from a story. They come out of two: the number a man publishes, and the number he says at his own table on a Sunday.

He looked at the caption again. *Asked our lab AI if we invented cold fusion. It said no and then begged us not to blow ourselves up. Best coworker.*

That was very good, he thought. That was better than good. If you had to get a thing seen and not believed, that was exactly how you would do it, and you would not do it yourself, you would have somebody young do it, somebody funny, somebody nobody would think to ask.

He got the napkin out of the shirt pocket and unfolded it on his knee and looked at his own handwriting, at *200 W*, and he did not cross it out.

Under it he wrote: *public number 200 mW*.

And under that: *both true*.

Then he opened the message board.

Chapter 4 — A Matter of Public Record

It took him four nights, and the first three were wrong, and that was the part he'd have wanted somebody to understand.

He did it in the garage after Nadine went up, at the bench under the good light, with the laptop propped on a coffee can because the fan was going and the fan needed air. He had a legal pad. He had the napkin flattened out and weighted with a socket. He worked as he had always worked, which was to write down what he actually knew and draw a line under it and not let anything past the line that hadn't earned it.

Above the line: *palladium. heavy water. calorimetry. Friday. "Danny," physicist, university, tenure or close. Two states off, maybe. Postdoc, young, funny, posts.*

Below the line, in the space he thought of as the tank — everything gets in the tank, most of it comes out garbage: *200 W. jar of water. free energy. AI safety system tracks everything.*

The two hundred watts was garbage. He'd known that on the stool. A salesman's number, rounded up out of a hole in his own knowledge, and if Dale had been the kind of man who needed a story to be clean he'd have thrown the whole night out on account of it. He wasn't. You didn't discard a casting because the sprue was ugly.

What was above the line was thin but it was hard, and hard was what you could work with.

He asked the message board first, because that was where he asked things.

It was a forum three clicks off the open internet, four hundred-odd members, run for eleven years by a man who signed himself ExcessHeatKev and who had written the long history of the 1989 replications that Dale had gone through twice. The board was where the people were who had never agreed to stop. It was also, most nights, four hundred people with one photograph among them, and Dale had watched it chase three false labs in the time he'd been reading it.

He posted what he had — palladium, heavy water, Friday, a physicist called Danny two states off — and went to bed.

hydride_h: two states off from where

dfurlong: from me

hydride_h: we don't know where you are

Verity: He isn't going to tell you where he is, and you oughtn't to ask him.

zoomerfusion: ran the bar. it's a franchise. there are forty of them

hydride_h: it's Texas. I'd put money on Texas

hydride_h: everything about it says Texas

Utah89: genuine question and I'm not being a jerk about it. has anybody considered that this is a joke. a postdoc said something funny to a chatbot on a Friday afternoon and a salesman in a bar heard about it fourth-hand

hydride_h: Utah.

Utah89: I'm asking.

Deuteronomy: Friends. Let us not begin by eating our own.

Sixty replies by morning and not one fact in any of them, and the only question in the thread worth answering was the one that got shouted at. A dozen of them were **hydride_h**, who had named himself after what palladium turns into when you fill it, and who Dale had liked from his first week on the board for exactly that reason.

So he closed it and did it himself.

He started with the postdoc, because the postdoc was the only actual person he could see. Her account was mostly jokes and a guitar and a softball team, and she had not once in three years said the name of the place she worked, which he noted with something close to approval. But in April she had posted a photograph of a whiteboard with a bad drawing on it, and behind the whiteboard, out of focus, was a window, and out the window was a water tower.

He spent most of Tuesday night on the water tower and got nowhere, and went to bed at two, and Nadine said something into the pillow that he answered with a hand on her back.

Wednesday he stopped being clever.

Because the thing about real work — actual funded work, the kind with an institution behind it — was that somebody had to write down what they were doing and somebody had to publish it, and that was not a leak or a slip or a thing you had to be smart to get. It was a rule. They took the public's money and in exchange they filed an abstract, and the abstracts sat in a database with a search box on it, free, indexed, open to any man with a library card and a reason.

He typed *palladium deuterium calorimetry* into the search box and got two hundred and eleven results, and he read them.

Not skimmed. Read. It took him until Thursday, and by the end of it he could have told you who in this country was doing what with a heavy-water cell and on whose dime, and if you had asked him a year ago whether he'd have found that interesting he'd have said no, and he'd have been wrong. It was the best three nights he'd had since the shop.

Most of them were nothing. Fuel cells, hydrogen storage, sensors, a man in Colorado doing something with membranes that Dale read twice for pleasure. Seven were in the neighborhood. Four were close.

And one of them was year two of four, at a state university two states east, with a principal investigator named Daniel Cho.

Dale read that name three times and then got up and walked to the end of the garage and back, and then sat down again and read the abstract.

This project supports a systematic calorimetric study of hydrogen and deuterium loading in palladium electrodes under controlled electrochemical conditions. The work will establish baseline measurement uncertainty for closed-cell isoperibolic calorimetry at the sub-watt level, and will characterize the contribution of instrumental drift to reported excess-power values. Outcomes are expected to inform reproducibility standards in low-energy electrochemical measurement.

He read it maybe seven times.

The first time, what he felt was the flat drop of disappointment, because there was nothing in it. No claim, no result, no fusion, nothing but the gray careful nothing that men write when they are asking for money and do not want to promise anything they might fail to deliver. It was four sentences long and it said, as far as he could tell, *we are going to check our thermometers*.

The second time, he noticed *excess-power values*.

He sat back.

There it was, in the public record, in a government database, under a man's own name. Not *excess power* as a thing they hoped to find. *Reported excess-power values* — other people's. Already reported. Sitting there as a known quantity that this man had been given four years and the public's money to *characterize the contribution of instrumental drift to*.

Which was to say: he was funded to explain them away.

Dale put the pencil down and looked at the wall for a while.

He was not a stupid man and he did not go fast. He made himself say the innocent version out loud, because that was the discipline — you say the boring answer first and see if it holds. The boring answer was that a fellow was doing careful measurement work on a phenomenon everybody had agreed thirty years ago was measurement error, and the abstract said so plainly, and there was nothing under it.

And that held for about a minute, and then it didn't, because of *low-energy*.

Nobody said low-energy. Not by accident. In thirty years of a field that had been laughed at under one name, the people still working in it had put a new name on the door — *low energy nuclear reactions*, and everybody who knew, knew, and everybody who didn't, didn't. And here was that phrase, filed down to *low-energy electrochemical measurement*, sanded to where a program officer's eye would slide right off it, sitting in the last sentence of a public abstract like a mark chalked on a gate.

You did not put that phrase in a document by accident. You put it in so the right man would find it.

He wrote on the legal pad: *Daniel Cho. Yr 2 of 4.* Then, because his hands wanted something to do, he squared the pad to the edge of the bench.

Then the postdoc's water tower, cross-checked against the campus in forty seconds now that he knew where to look, because the tower was right there in the third photograph on the university's own visitor page.

Friday. Palladium. Heavy water. Two states east. Daniel Cho.

He'd spent four nights searching. Three of them had turned up nothing.

He didn't post it that night.

That surprised him and he sat with it. He had it open — the reply box under Kev's pinned thread, the cursor going — and he had the name typed, and he did not push the key.

Because it was a man. That was all it was and it was enough to stop him for an hour. Somewhere two states east there was a fellow with a name and a wife probably and a job he'd gotten by being good at something, and Dale had spent four nights in a garage assembling him out of a database, and the second he pushed the key that man belonged to four hundred strangers.

He thought: *if I'm wrong, I've done something to a stranger for nothing.*

And then he thought: *if I'm right, and I sit here, then it's four years of public money going to bury the only thing that ever mattered, and I sat in a garage and let it.*

He looked at the napkin under the socket. *Both true.*

He pushed the key.

He was asleep in ten minutes and did not see any of it.

hydride_h: it's a name. it's an actual name

Utah89: ok everybody breathe. we have had four (4) of these. remember Rochester. remember the guy in Idaho who turned out to be a dentist

hydride_h: Utah this is not Rochester. go read it. he did the abstracts. all of them. he read two hundred abstracts

zoomerfusion: he read TWO HUNDRED AND ELEVEN abstracts

Utah89: ...I'm reading it

zoomerfusion: take your time

Utah89: ok that's the phrase. that's the phrase in the last line

hydride_h: THAT'S THE PHRASE

Verity: What I want on the record is the order he did it in. He didn't start from the lab and go looking for evidence. He started from a number a drunk got wrong and worked backward through the public record until it closed. Four nights, three of them dead ends, and he posted the dead ends.

Verity: Nobody here has ever done that. Including me.

Deuteronomy: He posted the dead ends.

Deuteronomy: Brothers, I have been on this board eleven years.

By the time Dale came down at six there were three hundred and forty replies and his phone had stopped being able to hold them all at once, and the thread had been pinned above Kev's, which had never happened to anybody, and somebody had already made a graphic out of the last line of the abstract with the words *LOW-ENERGY* boxed in red.

He stood in the kitchen in his socks and scrolled with one thumb and did not eat.

"You're up," Nadine said.

"I'm up."

"You were down here till two."

“Till one.”

“Till two, Dale.” She was doing the thing where she got the coffee going without looking at the coffee, because she was looking at him. “You’ve got to put Tuck’s new gasket in at nine.”

“I know it.”

“And the Ferreiras’ chest freezer.”

“I know it.”

She stood there a second. He could feel her taking the reading and he made himself put the phone face-down on the counter, which he understood, even as he did it, was the single most suspicious thing he could have done.

“Well,” she said. “You look better than you’ve looked in a year, so I’m not going to ask.”

“Nadine—”

“I said I’m not going to ask.” She handed him the cup. “Nine o’clock.”

He did Tuck’s gasket in fifty minutes and the door shut like a bank vault after and Tuck made him take an extra forty dollars.

Then a Buick came up his drive at ten with the trunk lid tied down over a push mower, and a woman got out of it backwards and in stages — sixty-some, built like a fence post, straw visor, a sleeveless blouse with hibiscus all over it.

“It doesn’t start.”

“Since when.”

“Two weeks. You pull on it till your arm gives out and it just sits there.”

He got the twine off the lid and walked the mower down onto the gravel and pulled it once and got nothing, and pulled it again and got nothing, and she watched it not start with a certain satisfaction, the way you watch a thing finally misbehave in front of a witness.

"It's the screen in the bowl," he said. "They plug up with whatever's in the bottom of your gas can. Buy your gas in May and don't buy it in April." He wiped his hands down his thighs, which did nothing. "Forty. If I get into it and it's more than forty you'll hear from me before I do the work."

She counted him out forty in fives, which was how she paid for everything, and snapped the purse shut on the rest.

"You just paid me for a guess."

"You'll be right."

"You'd think they'd build one that doesn't do that."

"They could."

She had the trunk lid halfway down, and Dale, who had been up till one three nights running with two hundred and eleven abstracts and a stranger's name in his head, heard himself say it before he had decided to.

"Can I ask you something. You ever think about what's actually in a glass of water?"

She stopped with the lid on her fingers.

"Not really."

"No."

"There's water in it, Dale."

"There is." He pushed the lid the rest of the way down for her, because that latch had never caught right and she'd never had it seen to, and she got back in backwards and in stages and went off down the drive.

That was the whole of what he said out loud to another person before two in the afternoon.

Then the Ferreiras' freezer, which was a relay, and while he had the truck in the drive Mrs. Ferreira walked him round to the dryer and stood there while it made the noise, and he ordered a belt for it that was going to take a month.

He was wiping down in the truck at two in the afternoon when the phone buzzed with something that wasn't the thread.

It was a private message, and it was from ExcessHeatKev, whose real name Dale did not know and had never wondered about, and who had run that board since before Dale found it and had personally written the twelve-part history of the '89 replications that Dale had read the way other men read scripture.

ExcessHeatKev — *private*

Dale. I've been at this a long time and I've watched a lot of people get excited.

You did something different and I want to say it plainly: you did the work. Nobody asked you to and nobody was going to check.

Now. There are eight or nine of us who are serious — actual benches, actual cells, people who have spent money. We talk somewhere else, not on the board. I'd like you in it.

One thing first, and I need you to hear it as respect and not as a test. You've got the lab. You've got the man. What you don't have is the log.

Everything you found is what they filed. I want what they said.

Dale sat in the truck in the Ferreiras' driveway with the engine off and the sun coming through the windshield onto his yellow fingertips, and read it twice, and understood that he had been offered something.

He typed back one line, because thirty years had taught him that the man who talks least at the table is the man they end up asking.

What they said to who?

The answer came before he had the truck in gear.

ExcessHeatKev — *private*

To the machine.

Chapter 5 — Clarification

The report went in on the eleventh of July, three days early, and for eight days Cho did not think about it once, which he would later identify as the last stretch of his life that had that quality.

Then Anne Sedlak called.

She was the program officer, and in two years she had called him twice, both times about a budget line, and she had the specific unhurried courtesy of a person whose job is to be pleasant to people whose careers she is holding.

“Daniel. Do you have a minute.”

“I’ve got several.”

“Good, because this is going to be stupid and I want to get it over with.” A chair moved somewhere. “Somebody sent me a picture.”

He knew, in the flat instant before she said the next thing, exactly which picture. He had not looked at it in nine days as a kind of hygiene, the way you don’t press a bruise.

“It’s my postdoc,” he said.

“I assumed. It’s funny. I laughed.” A pause with a shape to it. “It went to my director, Daniel, and my director does not know your name, and it arrived on his desk under a subject line somebody else wrote, and the subject line said *is this us*.”

“Is this us.”

“Meaning did this office fund it.”

“You funded a calorimetry baseline.”

“I know what I funded. I wrote the summary.” Sedlak sighed, and it was a real sigh, the sigh of a woman on her fourth conversation of the day about a thing that isn’t real. “Here’s where we are. He’s asked me to get a clarification in the file. Two sentences. Something I can point at.”

“A clarification of what? Nothing happened.”

“Of that. Of the nothing.” She said it kindly. “Daniel, I am on your side, and I want to be plain with you about how this works, because you’ve never had a bad year and you may not know. Nobody up there thinks you’ve discovered cold fusion. What they think is that there is now a document in the world with the words *cold fusion* and this office’s money in the same frame, and when that happens somebody senior wants a piece of paper that says we looked into it. That’s the whole thing. It is not an accusation. It’s a *file*.”

“And if I don’t write it?”

“Then there’s no piece of paper,” Sedlak said, “and the year-three release goes to a committee that has never heard of you, and they’ll ask for one anyway, in November.”

He looked at the bath, which was holding at 0.02, which was correct.

“Two sentences,” he said.

“Two sentences. Boring. Please make it boring.”

The first version took him four minutes.

No anomalous excess power has been observed in this project. Reports to the contrary originate from an informal exchange with a language model and do not reflect any experimental result.

He read it back and it was fine, and then he read it back a second time and it was not fine at all, because *reports to the contrary* was doing something he hadn’t asked it to. It admitted there were reports. It put them in the file. Anyone reading the sentence would learn, from the sentence itself, that reports existed — and would want to know what they were.

He cut *reports to the contrary*. Now it read:

No anomalous excess power has been observed in this project.

One sentence, true, complete. He looked at it for a while and felt the floor of it give.

Because they had asked for a clarification, and this was a denial, and a denial that doesn't say what it's denying reads like a man declining to answer a question you can now hear him not answering.

"Errol," he said. "Come look at this."

Whitfield came and read over his shoulder and made a small sound.

"You can't say that."

"It's true."

"It's not true, and that's my whole objection." He put a finger on the screen, which Cho hated and had told him he hated. "You *did* observe excess power. Two hundred and ten milliwatts of it, on a Friday, and I watched you write it down. What you did not observe is excess power *outside your measurement uncertainty*, which is a completely different sentence, and if you put that one in a federal file and somebody ever pulls your logbook you have handed them a discrepancy."

"Nobody is going to pull my logbook."

"Nobody is going to send your postdoc's joke to my director either," Whitfield said, and went back to his standards, having been right in a way that would take about a month to fully arrive.

Vega got up, crossed the room, and put a mark on the whiteboard without saying anything.

"That was not a *technically*," Whitfield said, to his standards.

"Forty-three," Vega said, and sat back down.

Cho sat there.

Then he wrote: *Apparent excess power of approximately 0.2 W was recorded during routine baseline runs and was subsequently attributed to bath drift. No excess power outside instrumental uncertainty has been observed in this project.*

It was accurate. It was the most accurate thing he could write. It was also, he understood, a document that now contained the phrase *excess power of approximately 0.2 W was recorded*, and he thought about a person reading

only the first eleven words of it, and how many people read only the first eleven words of anything, and Greg's voice at the patio table said *two hundred* with the units gone.

Vega rolled over. She had said one word in twenty minutes and it had been a number, and going quiet was not a thing Vega did.

"Is that about my thing."

"It's about a file."

"Dan."

"It's about a file," he said, "that exists because of your thing."

She read it. She read it twice, and the reading took the animation out of her face in a way he had not seen before and did not enjoy.

"I can take it down."

"It's got twelve thousand shares."

"I can still take it down. It just —" She stopped. She was working it out in real time and he watched her get to the end of it. "It doesn't go away. It just stops having me attached to it."

"That's about the size of it."

"Okay, then say it was a joke." She gestured at the screen. "That's the actual fact. It was a joke, it was Friday, I was being an idiot. Two sentences. A *postdoc made a joke*. Done."

"I can't put the word *joke* in this."

"Why not, it's the truth—"

"Because it's a federal file and I have to say the thing that's true in ten years, not the thing that's true today." He rubbed his eyes under the reading glasses. "And because nobody has ever gotten out of anything by writing down that it was a joke."

Vega sat with that.

"You know what'd be easier?" she said. "Having actually done it. That's one sentence. You'd be at lunch."

He laughed before he could help it, and it was the only funny thing that had happened all week.

He almost sent it. Then he pasted it into the gray box, because he ran everything through the gray box and had for a year, and it had never once failed to earn the four seconds.

read this. it's going in a federal file. i want it accurate and i want it boring.

It's accurate, and it's appropriately dull. One flag: it opens with "Apparent excess power of approximately 0.2 W was recorded," and a reader who stops after the first clause — which readers do — comes away with the impression that excess power was recorded. If you want it hard to misquote, lead with the conclusion instead. "No excess power outside instrumental uncertainty has been observed in this project. An apparent 0.2 W signal recorded during routine baseline runs was attributed to bath drift." Same facts, same accuracy, and the first eleven words say the thing you want said.

For what it's worth, neither version is wrong. This is a question about how it will be read, not about whether it's true.

Cho looked at that for a while.

It was better. He knew it by the second sentence. And he did not use it, because somewhere in a drawer eight feet away was a paper logbook with *two-ten* in his own hand on a Friday in June, and if a man ever pulled that book and then pulled this file, Cho wanted the two-ten sitting in the first line of it. The machine was right that leading with the number made it easy to misquote. But a misquote he could answer. A man asking why he had moved the number further down the page he could not.

So he kept his own version, for a reason he could have defended to anyone, and sent it.

Sedlak wrote back inside the hour with *perfect, thank you, boring is exactly right, it's in the file*, and for about thirty hours he believed that was the end of it.

On the Thursday, the university's records office sent him a form.

He almost didn't open it. It came from an address he had never had cause to notice and the subject line was a case number, and he opened it standing up with his bag still on his shoulder at ten past eight in the morning.

This office has received a request under the state Public Records Act. You are receiving this notice because responsive material may be in your custody. Please do not delete, alter, or archive any material described below pending our determination. A response is due within twenty business days.

He read the description twice.

All logs, transcripts, session records, prompts, outputs, and related communications generated by any artificial intelligence or machine learning system licensed to the University, for the period beginning June 1, associated with the laboratory of Dr. Daniel Cho, Department of Physics; and all correspondence referencing excess heat, cold fusion, or low energy nuclear reactions among the same.

There was a name at the bottom, in the box where the requester's name goes.

D. Furlong.

It meant nothing to him. It was not a newspaper and it was not a law firm, and the line under it that said *Organization* had been left empty, and the line under that was a street address in a town he had never heard of, two states away. In the box marked *Requester type*, somebody had checked **Individual**, and beside it in small print the form explained that individual requesters were not assessed commercial-use fees.

Cho stood in the doorway of his own lab with his bag still on his shoulder and read the phrase *please do not delete* three times, and it occurred to him, with the particular clarity of a thing arriving twelve days late, that he had never once in his life had a reason to wonder where the Sage sessions went.

Chapter 6 — Twenty Business Days

The statute ran to nine pages and he read all nine at the kitchen table on a Sunday with a cup going cold at his elbow, and what surprised him was how friendly it was.

He had expected a wall. He had expected the kind of document that exists to make you give up partway down the second page, because that had been his experience of the written word in an official font — the unemployment forms, the shop's closure notices, the letter about the pension that had taken four readings to reveal that it said no. This was not that. This said, in plain sentences, that the records of a public body belong to the public, that any person may request them, and that the body shall respond within twenty business days.

Any person. It said it twice. It did not say a lawyer, it did not say a newspaper, it did not say a person with a reason. Nobody was required to want it for a good cause. There was a paragraph explaining that a requester need not state a purpose and that a body may not ask.

At the bottom of the page the state had put a link to a form, and beside the link, a note that individual requesters were not charged the commercial fee.

It cost him nothing. That was the part he came back to, later, more than once. There was no gate. He had spent a year assuming that a man like him was on the outside of things by some arrangement nobody would ever explain to him, and here was the arrangement, published, indexed, with a form at the bottom.

He filled it in.

He was careful with the wording, because thirty years of writing parts orders had taught him that you get what you ask for and not what you meant. He did not write *cold fusion logs*. He wrote out the whole thing, every category he could think of and then the ones Kev suggested — *logs, transcripts, session records, prompts, outputs, and related communications generated by any artificial intelligence or machine learning system licensed to the University* — and he put a start date on it, June 1, which was early enough to be safe and recent enough not to look greedy.

Then the requester box.

He sat with that one, and afterward he could not have told you why for more than a second, because the answer had been obvious. He typed *D. Furlong* and his street address, and left the organization line blank, because there wasn't one.

He was not sneaking. He wanted that on the record and there was nobody to say it to. A man who was doing something wrong would have used a lawyer or a friend's name or nothing at all, and he had considered exactly none of those, because it had not once crossed his mind that he was doing anything but what the nine pages said he was entitled to do.

He hit submit at 4:14 on a Sunday afternoon and got a case number back inside a minute, and he wrote the case number on the legal pad under *Daniel Cho. Yr 2 of 4*, and then he went out and put the belt on the Ferreiras' dryer, which had taken a month to come and took seven minutes to fit.

He told Kev that night. Kev had asked him for the log four weeks earlier, and this was the only answer Dale had, so he sent it plain: *I asked the university for it. There's a state law. They have twenty business days to tell me yes or no.*

ExcessHeatKev — *private*

You did what

ExcessHeatKev — *private*

Dale. Dale. Eleven years. Eleven years of people telling me it can't be gotten.

Nobody ever asked them for it.

They put him in the room that night.

It was eight people, and it was not the board. The board was four hundred members and a public thread and everybody talking at once. This was a private chat with eight names in it, and nobody in it said anything unless they had something. Kev. Verity, who turned out to write like she talked and

to be the only one who ever said *I don't know*. A man in Alberta called Sethi who had built four cells and destroyed three of them and would tell you exactly how. Two brothers in Michigan who ran a machine shop and posted off one handle between them. A retired refinery instrument tech who went by rwhitcomb and who had spent thirty years being the man they called when a gauge disagreed with a gauge.

And Deuteronomy, who did not build anything, and who welcomed him with a paragraph so warm and so long that Dale read it twice and felt his ears go hot.

The first hour was the best hour he'd had in years.

Sethi: what are you running for a supply

dfurlong: bench rectifier. pulled it out of the shop's scrap when they drained the tanks

Sethi: constant current or constant voltage

dfurlong: both. I ran nickel off it twelve years

Sethi: oh good. you're not a hobbyist

rwhitcomb: he's real

That was all it was, six lines, and it went through Dale like a drink. Nobody had asked him a competent question in fourteen months.

Then the twenty days.

ExcessHeatKev: ok. we should plan for what comes back

Sethi: plan for nothing coming back. nothing is what comes back

ExcessHeatKev: they have to answer. it's in the statute

Sethi: they have to answer. they don't have to answer with anything

Verity: You're both wrong in the same direction. Whatever they send is the shape of what they're keeping, even when it isn't the thing itself. A flat no tells you something. A partial tells you a great deal more.

Verity: Dale. When it comes — don't read it by yourself the first time.

dfurlong: why not

Verity: Because the first read is the one you'll believe.

He said he wasn't going to sit a month waiting on a mailman.

dfurlong: I'll build while we wait

tolliver_machine: build what

dfurlong: a cell. same rig they've got. palladium wire down the middle, heavy water, insulated jacket, thermocouple in the cap

dfurlong: it's not exotic. it's a beaker and a wire and a way to measure the heat coming off it. I've got the rectifier already

Nobody typed for four seconds, which in that room was a long time.

Verity: With what money?

Palladium wire, four nines pure, one millimeter, came to eleven hundred and forty dollars for a length he could have held in a closed fist.

He knew the number before he looked it up. He'd have known it within fifty dollars — he'd priced anodes for a living — and knowing it in advance was not the same as watching the total sit there on the screen with a shipping estimate under it.

Heavy water was cheap, which was worse. The price of a thing is where you stop and think, and there was nothing here to stop at: it was ordered before he had finished deciding whether to order it.

He had four thousand and change in the account that was his, from the shop's last payout, that Nadine had never once suggested he do anything with. He did the arithmetic on the legal pad, and then he did it again on the other side, which changed nothing.

He was still doing it when she came in from the clinic and put her bag down and said, "There's a charge."

"There's a charge."

"Eleven forty." She got her shoes off, one heel against the other, and did not sit. "I saw it come through on the phone while I was on hold with a pharmacy. Eleven hundred and forty dollars, to a metals company in New Jersey."

"Yes."

"For what."

"Wire."

Nadine stood in the kitchen doorway with one shoe in her hand and looked at him for what he would later think of as a long time and what was probably three seconds.

"What kind of wire is eleven hundred dollars."

"Palladium."

"Is that for a job?"

"No."

He watched her decide not to ask the next question. He knew the shape of her doing it, the way you know a machine's sounds — the small settling, the hand going flat on the counter. She had a word available to her. She had had it available for eleven years and had never once used it, and he understood in that moment that she was not going to use it now either, and that this was not mercy, it was something more like a rule she had made about herself a long time ago and had kept.

"It's my money," he said, which he had not meant to say and which came out sounding exactly as bad as it was.

"It is," she said. "That's right. It is."

She put the shoe down and got a glass and filled it at the tap and stood with her back to him drinking it, which was his own move, and he recognized it being used on him and could not think of one thing to do about it.

“Are you going to tell me what it’s for?”

“I’d have to start a long way back.”

“I’ve got till Thursday.”

He almost did it. He got as far as *there’s a lab* and could see the whole thing lined up behind it — the salesman, the abstract, the phrase in the last line, the machine saying please — and he could hear how it sounded in his own kitchen, in his own voice, out loud, to her.

“It’s an experiment,” he said. “It’s real chemistry. It’s the same chemistry I did for thirty years, just smaller.”

“Okay.”

“It’s not —” He stopped. “It’s not a thing somebody sold me.”

“Okay, Dale.”

She said it without any edge on it at all, and that was the thing that got him, because an edge he could have argued with. She rinsed the glass and set it upside down on the rack and went to get out of her clinic clothes, and at the doorway she stopped.

“Is anybody getting hurt by this?”

“No.”

“All right,” Nadine said, and went upstairs.

He built in the garage, nights, and it went well.

That was the thing nobody who has not done it understands. It went *well*.

The bench was clean and the light was good and his hands knew what they were doing before he asked them. He cut the jacket. He drilled the cap for the vent and the thermocouple on the drill press with the piece clamped, and he drilled it true, and when he held it up against the light the two holes

were where he had said they would be, which is the whole of that particular pleasure.

The wire was the good part. Four nines palladium, one millimeter, and it came in a plastic sleeve inside a box big enough for a shoe, and he had priced anodes for thirty years and had still never had metal in his hand that cost what that cost. He turned a form out of a dowel and wound it on slow. He had wound harnesses in a tank room with his gloves half off and a foreman two feet behind him for thirty years, and it turned out that he had never once had a job where nobody was standing there, and it turned out that he liked it.

The first cell leaked. It leaked at the thermocouple gland, about a bead a night, slow enough that he lost two evenings deciding whether he was imagining it. He found it at half past ten with dish soap and a flashlight — one bubble swelling and letting go and swelling again — and he sat back on the stool and looked at it, and he was happier at that moment than he had been at any point since the Tuesday the envelopes came.

He cut a new gland with a shoulder on it and a fiber washer under the nut and the second one held dry for two days and wet for two days after that, and he wrote both down.

Arnie came down from up the road on the Sunday, bandy-legged, the white band already across his forehead where the cap sat, carrying a coffee can of bolts he was sorting for no reason anybody had asked him to.

He stood at the mouth of the garage and looked at the bench a while.

“That’s not a compressor.”

“It’s not.”

“That’s glass.”

“It is.”

Arnie got two bolts out of the can and held them against each other and put the shorter one back. He came in three steps, which was as far as Arnie

ever came into anybody's garage, and looked at the tubing running off the cap and out into the room.

"Where's that go."

"Out."

"Out where."

"Just out. It's got to breathe or it builds up."

"What builds up."

"Gas," Dale said. "Same thing that's come off the battery in your truck for forty years. You run current through water and it comes apart on you."

"Huh," Arnie said, and did not ask anything further about it, because he had been given an answer and the answer had a truck battery in it and that was as far as you ought to go on a Sunday.

He put the can against his hip. He was in no hurry to go and he did not appear to be waiting for anything either.

"What's it for?"

"It's an experiment."

"That's not an answer, that's a category."

"It's the answer you're getting."

"Fair enough," Arnie said, and did not seem troubled by it in the slightest, and stood there another minute looking at the cap. "That's nice work on that lid."

"It's all right."

"It's better than all right and you know it. You've been stood there ten minutes waiting on somebody to say so."

"I have not."

"You've got it turned toward the door."

Dale looked at the cell, which was turned toward the door.

Arnie went off up the road and the sorting in the can started again about the time he got past the mailbox, and Dale stood in his own garage and understood that he had just had the best four minutes of his week off a man he would not have crossed the road for in 1990.

Sethi walked him through the calorimetry over four evenings and was a good teacher and told him twice that his own numbers had been garbage for a year before he'd figured out why, and Dale liked him for that.

And the cell sat on the bench under the good light and made gas, and vented it out a length of tubing into the room.

Twenty business days is a month. He marked them off on the legal pad, one stroke a day, four upright and the fifth laid across them, and did not look at the strokes more than twice an evening.

On the nineteenth day an envelope came from the university, and it was thick, and Dale carried it into the garage and put it on the bench under the light and did not open it, because Verity had said don't read it alone the first time and he had said he wouldn't.

He sat and looked at it for twelve minutes.

Then he opened it.

Chapter 7 — Exempt from Disclosure

Denise Kovach from the Office of Public Records came to the lab on a Tuesday with a laptop, a rolling bag, and no sign at all of finding any of it unusual.

“This is going to be boring,” she said, shaking his hand. “That’s the good outcome. Where can I sit?”

He gave her Vega’s bench. Vega was at a seminar. Whitfield was at the far end reshelving, ostentatiously not listening, which Cho could hear him doing from thirty feet.

“Walk me through what you’re worried about,” Kovach said, unpacking.

“I’m not worried.”

“Everyone says that and then at about the ninety-minute mark somebody remembers an email.” She got the laptop open. “I’d rather hear it now. It’s much easier for me at hour one than at hour eight.”

Cho sat down on the stool by the calorimeter.

“There’s a session where my postdoc asked the system whether we’d discovered cold fusion,” he said. “As a joke. It said *no* four different ways and then told us not to seal a cell. Somebody screenshotted it and it went around.”

“I saw it. My sister sent it to me.” Kovach typed. “It’s funny.”

“It’s cost me about three weeks.”

“Sure.” She was not being unkind. She was doing intake. “Anything else.”

He thought about the drafting session, and about how to say it, and there was no way to say it that took fewer than four sentences, and every one of the four made it sound worse than the one before.

“I asked the system to review a clarification I wrote for my program officer,” he said. “About the joke. Whether the wording was accurate.”

Kovach’s hands stopped for the first time.

“How many drafts?”

“Three.”

“Are they all in there?”

“I assume everything’s in there.”

“Everything’s in there,” she agreed, and typed for a while, and did not look up. “Okay. That one we’ll want to look at carefully.”

It took ten hours over two days and most of it was Cho sitting on a stool watching a woman read.

The lab’s Sage account had four hundred and sixty sessions in the request window. Kovach opened every one of them. Most were nothing — a units conversion, a table reformatted, Vega asking it to explain a supplier’s spec sheet, Whitfield asking it a question about the 1926 literature and then arguing with the answer for forty minutes.

Cho read over her shoulder for a while and then stopped, because it was doing something to him he hadn’t expected.

He had never once looked at these. That was the thing. He used the machine the way he used the sink. Two years of it, four hundred and sixty sessions, and every one of them was him and his people thinking out loud in a room they’d believed was empty — half-formed, sloppy, wrong on the way to right, saying *what if* about things they didn’t mean and abandoning them mid-sentence.

None of it was damaging. All of it was undignified.

“This is normal,” Kovach said, without turning around. “Everybody hates this part. You’ve never seen yourself from outside before and now you have and it’s not flattering. It doesn’t mean anything’s wrong.”

“How can you tell what I’m doing back here.”

“You went quiet.” She kept reading. “The quiet ones are the ones who are fine.”

On the second afternoon she turned the laptop around.

“Here’s where we land. I’ll write it up properly but I want you to hear it from me first, because you’re going to have opinions and I’d rather have them now.”

“All right.”

“The cold fusion session goes out. All of it. Both the prompts and the responses.”

“It’s a joke.”

“It’s a record created on a university system in the course of university business, and there’s no exemption that touches it. It isn’t personnel, it isn’t a student record, it isn’t attorney work product, and it isn’t research data.” She tapped the screen. “It’s three people being funny at four o’clock on a Friday, which is not a category the legislature made an exception for.”

“And the rest of it?”

“Most of the rest of it is research data.” Kovach scrolled. “Your state has an exemption for unpublished research data at public institutions. It’s there to keep people from filing requests to scoop your work before you can publish it, and it is broad, and it applies to essentially everything in here that touches your actual project. Your loading runs. Your calorimetry. The baseline analysis. All of that comes out.”

Cho looked at the screen for a while.

“So you’re going to release the joke,” he said, “and withhold the science.”

“I’m going to release the record that isn’t exempt and withhold the records that are.”

“Denise.”

“I hear it.” She did, too. She sat back and took her glasses off and rubbed the bridge of her nose, and for a second she was a person and not an office. “Believe me, I hear it. It is a genuinely stupid outcome and I don’t have a lever for it. The exemptions are the exemptions. I can’t release your research data because it would be convenient, and I can’t withhold a joke because it’s embarrassing.”

“Can I waive it? The research data. Can I just — release everything?”

She put the glasses back on.

“You could. It’s your data and the exemption exists for your benefit, so you can waive it.” A pause. “I’d advise you not to, and I’ll tell you exactly why, and then it’s your call and I’ll do whatever you say. If you waive an exemption on a request like this one, the next requester cites it. And there will be a next requester, because these come in waves, and the second one always asks for more than the first. Right now you have a shield you didn’t ask for. You want to think hard before you put it down.”

“And the clarification drafts?”

“Those go out.”

“Under what theory?”

“Under the theory that they’re not research data and they’re not privileged. You wrote a statement for a federal sponsor and you asked a tool to check the wording.” Kovach shrugged, not unsympathetically. “It’s the least interesting document in the file. It’s a man proofreading.”

“It’s a man proofreading a statement about cold fusion.”

“Yes,” she said. “It is that.”

He signed the release Thursday morning and the packet went out Thursday afternoon: forty-one pages released in full, a hundred and six items withheld, and a schedule at the front listing every withheld item by date and description with the statutory citation printed beside it in a column of its own.

Before he signed he did one other thing, which took four minutes and which he did not mention to anybody.

He typed the requester’s name into a search box.

There was almost nothing. A *Furlong, D.* on a fifteen-year-old high school reunion page. A business listing for a plating shop in the same town, marked *permanently closed*. A photograph, from a local paper’s archive, of eleven

men in front of a loading dock under a banner about a safety milestone, the caption running left to right with nine names in it and two unidentified.

Cho looked at the photograph for a while and could not tell which one he was.

Then he closed it, because there was nothing there, and because a man was allowed to file a form.

The envelope came on the nineteenth day and Dale opened it alone in the garage under the good light, and the first thing that fell out was the schedule.

He hadn't expected a schedule. He'd expected either a stack of paper or a letter saying no.

What he had was three pages, in a table, listing every single thing they were not giving him. Date. Description. Reason.

07/14 — Session record, calorimetric analysis. WITHHELD. §4-9-102(b)(11), unpublished research data.

07/14 — Session record, loading protocol discussion. WITHHELD. §4-9-102(b)(11).

07/19 — Session record, baseline drift characterization. WITHHELD. §4-9-102(b)(11).

It went on. He counted them later, twice, and got a hundred and six both times.

And behind the schedule, released in full, unredacted, every word of it, was the conversation from the sixth of June in which a postdoc asked whether they had discovered cold fusion and a machine told her four separate times that they had not, and then told them not to seal the cell.

Dale sat with the two halves of it on the bench in front of him, the schedule on the left and the forty-one pages on the right.

He was very calm. He noticed that about himself and it was true.

They had given him the joke.

They had given him the whole joke, every line, the Stockholm bit, the *off the record* bit, the *please don't seal it*, all of it, cheerfully, complete — the one document in the building that the entire internet had already seen, that his own board had had since June, that was on a t-shirt somewhere by now. They had handed it over in full and put a cover letter on it.

And beside it, in a column, with a law printed next to each line, was everything else. A hundred and six times. *Loading protocol discussion. Calorimetric analysis. Baseline drift characterization.*

Withheld.

He got up and walked to the end of the garage and back, as he had the night he found the name.

Because here was the thing, and it was so plain that he said it out loud to nobody in the empty garage: **if the joke was the whole story, there was nothing to keep.**

If a postdoc had made a gag on a Friday and the number had been the room warming up, then there was no research data. There was nothing to protect. A man with a null result has never in the history of the world needed a statute to keep it quiet.

They had not said no. That was the part that undid him. A flat no he could have fought or filed again or gone around. What they had done was hand him a receipt — a signed, dated, itemized, legally-cited receipt for one hundred and six documents that exist, that are about loading and calorimetry and drift, that are in a building two states east, and that he was not going to see.

Verity had told him not to read it alone the first time, and he had, and he understood now what she'd meant and that she had been right and that it was too late, because the first read was the one he was going to believe.

He looked at the schedule for a long time. Then he photographed all three pages and put them in the room.

dfurlong: they answered

dfurlong: [3 images]

Sethi: that's a denial

dfurlong: it's not a denial. read it

Sethi: 106 withheld. that's a denial with extra steps

rwhitcomb: no. no it is not. Sethi look at the middle column

rwhitcomb: they told him what each one is ABOUT. they didn't have to do that. I've written these. you can say "records responsive to your request" and be done

rwhitcomb: somebody in that office listed a hundred and six documents by subject matter and put a law next to each one

ExcessHeatKev: loading protocol discussion

ExcessHeatKev: loading protocol DISCUSSION

tolliver_machine: what's the joke doing in there

dfurlong: released in full. all of it. the whole thing everybody already has

tolliver_machine: they gave you the thing on the tshirt

dfurlong: they gave me the thing on the tshirt

It went on like that for two hours and Dale answered about a third of it and read the rest.

Then Verity, who had not said anything.

Verity: Dale. Did you read it by yourself?

dfurlong: yes

Verity: All right.

Verity: I'm going to say the boring thing once and then I'll stop. There is a version of this where a records officer applied a broad exemption as she applies it every week, and did not think about it for one minute,

and the reason the joke came out is that a joke isn't research data. That version explains every single thing on those three pages and it requires nobody to be hiding anything.

dfurlong: I know

Verity: Do you.

dfurlong: I said it out loud in the garage before I said anything to anybody. I say the boring one first. I've always said the boring one first.

Verity: And?

dfurlong: and a man with nothing doesn't need a law to keep it

There was a long pause in the room and then Deuteronomy filled four screens, and Dale scrolled past it, and later felt bad about that.

ExcessHeatKev: so we appeal

rwhitcomb: you can appeal. it goes to the same office. it takes ninety days and they cite the same statute

Sethi: so that's it then

ExcessHeatKev: that's not it

Sethi: Kev it is very much it. we asked. they said no. politely, in a table, with a law

dfurlong: they said no to the form

Dale put the phone down on the bench and did not pick it back up for a while.

He got up and stood over the cell — his own, wound by hand, drilled true, the vent line running quietly out into the room — and thought about a building he had never seen with a water tower behind it.

Because the form had taken him exactly as far as a form was ever going to take him. He had used it correctly and it had worked correctly and what it had produced was a receipt.

There were a hundred and six sessions he was not allowed to read. They were on a server in a building. And buildings are open on weekdays.

Chapter 8 — Released in Full

The thing Cho had not understood, and that took him about nine days to understand, was that the packet made it worse.

Not neutral. Not embarrassing. Worse — and worse in a specific direction he could not have predicted and could not now do anything about, because the joke that had been going around all summer as a joke was now going around as **a document produced by the University in response to a lawful request**, with a cover letter, a case number, and a date.

Somebody had photographed the cover letter and it was everywhere. The letterhead did it. A summer of *ha ha look at the polite robot* had been one thing. A university seal at the top of the same page was another, and Cho, who had spent his professional life in a building with that seal on the door and had never once thought about it, learned in nine days exactly what it was worth to people outside.

“It’s authenticated now,” Vega said. She had the thread open and had gone the color of the bench. “That’s the word they’re using. *Authenticated*.”

“It was always authentic. It’s a screenshot of a thing that happened.”

“Right, but now it’s *stamped*.” She scrolled. “Somebody made a — Dan, somebody made a side-by-side. My post on the left and the university’s version on the right, and the caption is *identical to the character, so what else in the packet is real*.”

“Everything in the packet is real. That’s what a packet is.”

“I know. I know that.” She put the phone face-down and then immediately turned it back over.

The email started on the Friday.

He got two hundred and some over the next week and read most of them, which he later concluded was the mistake, and they came in about five flavors.

There were the physicists, who were the worst. Not hostile — worse than hostile. Amused. Four separate people he had published alongside sent

him some version of *saw your name trending, hope the Nobel committee is treating you well*, and one of them, a man Cho had liked for fourteen years, appended a photograph of a whiteboard on which somebody at his institution had written **CHO CONFIRMED?** with three question marks after it, as though this were the funniest thing in the world, which for that man on that Tuesday it was.

There were the reporters, two, both polite, both explaining that they were not writing anything yet.

There were the people who wanted to give him money. Six of those. One offered forty thousand dollars for a “consulting relationship” and included a term sheet.

There were the hostile ones, which were the easiest, because there were only about seven of them and they all had the same three sentences in a different order.

And then there was the last kind, which he did not have a name for and which he stopped being able to read after the fourth one.

Dr. Cho — I dont expect an answer. I have followed this since 1990. I am 71. My wife thinks I am foolish and she is probably right. If it is nothing I would rather hear it from you than from anybody else. Thank you for your time either way.
— R.

He wrote back to that one. It took him fifty minutes and four drafts and he told the man, plainly and without condescension, that there was nothing, that the number had been the room warming up in the afternoon, and that he was sorry.

The reply came in under an hour.

Thank you for confirming that you have to say that. I understand the position you are in. Good luck to you and yours. — R.

Cho sat and looked at that for a long time.

“Errol,” he said finally. “Come read this.”

Whitfield read it over his shoulder, and for once in his life had nothing to say for about ten seconds.

“Ah,” he said. “Yes. There’s no move, is there.”

“There’s no move.”

“If you’d ignored him he’d have taken that as confirmation, and you answered him and he took it as confirmation, and if you’d sent him the raw calorimetry he’d have gone down it looking for the seam where you took the real data out.” Whitfield straightened up. “It’s not a proposition, Daniel. It’s a *shape*. Everything you put into it comes out the same side.”

“That’s very helpful.”

“I didn’t say it was helpful. I said it was true. You know I’ve never confused those.”

Ellen called on the Tuesday, in the middle of all of it, about a Saturday.

“I need a date.”

“For what?”

“For the thing we do every year, Danny. The one where people come to our house and eat.” Something went down on a counter at her end; she cooked and called, she had always cooked and called. “Dawn asked me at church. Twice. I put her off both times and I’m not doing it a third time, because at three it stops being a scheduling problem and starts being a statement.”

“Can we look at it again in a couple of weeks?”

“That’s what you said in July.”

He looked at his inbox, which had sixty-one unread in it since lunch.

“The twentieth,” he said.

“You don’t mean that.”

“I said the twentieth.”

“You said it in the voice.” She let a pause go by with a wooden spoon in it. “Danny. If we do this, my brother is going to be standing in our kitchen for five hours.”

“I know.”

“He is going to tell eight people about the cold fusion.”

“I know he is.”

“And you’re going to stand there holding a beer and let him, because you can’t correct him in front of Dawn Petrosky without it turning into the whole evening.”

Cho sat back. Down at the far bench Whitfield was having an argument with a spectrophotometer.

“Book the twentieth,” he said.

“I’m booking it.”

“Book it.”

“It’s booked,” Ellen said. “And if you cancel I’m having it anyway and telling everybody you’re at a conference.”

Sedlak called on the Wednesday and did not open with a joke.

“How bad is it,” Cho said.

“It’s not bad. I want to say that in the first sentence because I know you’ll be counting them.” Paper moved. “Here’s the actual situation. Your year-three release is on the schedule for the ninth of November. Nothing has changed about that. Your work is good, your reporting is clean, and the file has the clarification in it and the clarification is exactly right.”

“But.”

“But the request came in through our office too.”

Cho closed his eyes.

“Federal, this time,” Sedlak said. “Different requester, different statute, and I can’t tell you who because I genuinely don’t know — it comes through a queue. They’ve asked for all correspondence between this office and your institution referencing excess heat or cold fusion, which is a category that now includes a phone call I made to you and a two-sentence clarification I asked you to write.”

“So your file is in it.”

“My file is in it. Which is fine, Daniel, because my file is a program officer doing her job correctly, and that is what it will look like to anyone who reads it.” A pause. “It will also look like a federal agency corresponded internally about cold fusion, four times, in twelve days.”

“That’s not what happened.”

“That is precisely what happened,” Sedlak said. “That’s the trouble with it.”

He went back to the lab and stood in the middle of it for a while, and then he did the thing he had been not doing for two weeks, which was ask.

He pulled the thread into the gray box — the whole thing, four hundred posts, the side-by-side, the schedule with the hundred and six lines, the graphic with *LOW-ENERGY* boxed in red — and typed.

explain these people to me. I’m not being rhetorical. I want to understand what they actually believe and why nothing I do lands.

I’ll do my best, and I’ll be direct rather than diplomatic, since you asked.

They believe a real anomalous result was obtained in your lab and is being suppressed. What holds that belief together isn’t any single piece of evidence — it’s that the belief is structured so that every possible observation supports it. Consider what your available actions produce:

If you say nothing, silence is consistent with suppression. If you explain, an explanation is what a person under instruction would provide. If you release the underlying data, the data can be read as sanitized, and its cleanness becomes evidence of preparation. If you refuse to release it, the refusal is the clearest evidence of all.

This isn’t a failure of your explanations. A claim that is compatible with every outcome cannot be reduced by producing outcomes. There is no experiment you can run on them.

Two specific things are critical right now, and I’d flag both. The exemption schedule is the strongest artifact they have, because it is an official

document that enumerates withheld records by subject. And the safety line — “please don’t seal it” — is functioning as the center of it, because a warning is read as knowledge of a danger, and knowledge of a danger is read as knowledge of a mechanism.

Cho read that twice.

so what do I do.

Honestly? Very little that’s satisfying.

You can keep your own record clean and accurate, which you have. You can decline to produce new artifacts, because each new document becomes new material to interpret — this is the main practical lever you have, and it argues for saying less rather than more. You can let it decay, because most of these lose energy in a few months when nothing further happens.

What I would not do is try to win it. Every attempt to close it has so far enlarged it, and that pattern is not going to reverse.

I’m sorry. I realize that’s not much.

“What’s it say,” Vega said from across the room.

“It says there’s nothing I can do.”

“Great. Cool. Very helpful machine.”

“No,” Cho said, and he was surprised by how flat his own voice came out. “It’s right. It’s the only thing in nine days that’s been exactly right.”

She deleted the account on the Thursday.

He found out because Whitfield mentioned it — Whitfield, who did not have a phone that did anything and who had learned it from a graduate student in the hallway, and who reported it to Cho the way he reported a units error, as a fact requiring attention.

“All of it?”

“The entire thing, apparently. Three years of it. The guitar as well.”

Cho went and found her in the cold room, where she had gone to inventory standards that did not need inventorying, and stood in the doorway.

“You didn’t have to do that.”

“I know.”

“Mari.”

“I know, Dan.” She had a clipboard and was not writing on it. “It wasn’t a gesture. I want to be clear that it wasn’t a gesture, because if it was a gesture then I did it for me and I’ve been trying pretty hard for two weeks not to do anything for me.”

“Then why.”

“Because they were reading it.” She said it evenly, which was the thing that got him — Vega never said anything evenly. “Not the post. The rest of it. Somebody made a timeline of my last eighteen months and put it next to the runs. There’s a picture of me at a wedding in March and there’s a guy explaining what it means that I look tired.”

Cho did not have anything for that.

“So it’s gone, and I feel about nine percent better, and I’ve worked out that the nine percent is the part where I stop being the one they’re looking at, which is exactly the thing I said I wasn’t doing.” She set the clipboard down. “Anyway. I want to post a correction.”

“Post it where? You deleted the account.”

“I’d make a new one. One post. Not funny. Just — here is the number, here is what the number was, here is why it was nothing, signed, the actual person who asked the actual question.” She had clearly written it already, several times. “I’m the only one of us they’d believe, Dan, because I’m the one who *started* it. It’s the one thing I’ve got.”

“No.”

“You haven’t heard it.”

“I don’t need to hear it. It’s good. I’m sure it’s good.” He leaned on the door frame. “Mari, four hours ago I asked that thing what to do and it told me the single practical lever I have is to stop making documents. Every new thing we produce is a new thing to read. Yours would be the best one they’ve had yet, because it’s from you, and it’s sincere, and it’s got a number in it.”

“So we just take it.”

“We just take it.”

“That’s a garbage answer.”

“It’s the only one I’ve got, and I’ve been looking.”

She picked the clipboard back up and looked at it and put it down again.

“I keep thinking about the guy,” she said. “Your seventy-one-year-old. R.”

“Don’t.”

“He’s got a wife who thinks he’s foolish. He said so in the email. He wrote that to a stranger.” She rubbed her face with both hands. “I made a joke on a Friday, Dan, and somewhere there’s a woman having an argument with her husband about it.”

Whitfield’s voice came from somewhere behind Cho in the corridor, not raised, not coming closer.

“She has been having that argument since 1990,” he said. “You are not that important, and neither is he. Come out of the cold room, both of you, it’s four degrees in there and I’ve made the bad coffee.”

Whitfield’s turn came on the Friday, and Cho would remember it as the only funny thing that happened that month.

It arrived as a link, sent by a former student with the subject line *you need to see this immediately*, and it was a post on the board, four thousand words long, titled *WHITFIELD: THE MAN IN THE ROOM*.

Somebody had gone through the released packet — the whole forty-one pages — and noticed that the sealing question had not come from the postdoc and had not come from Cho. It had come from a third person,

unnamed in the transcript, referred to once as *errol*. And they had found him: emeritus, thirty-one years, retired, back in the building unpaid, present in the lab on a Friday afternoon with no obligation to be there.

The post laid it out with real care. Why would a retired man come in for free? Why is the only technically substantive question in the entire log his? Why is he not on the grant?

He isn't on the grant because he doesn't need to be, the post concluded. *Cho is the name on the paperwork. Whitfield is the man in the room.*

Vega laughed until she had to sit down on the floor.

"You're the mastermind," she said. "Errol. Errol. You're the *shadow* mastermind."

Whitfield had read the whole thing standing up, with his glasses down his nose, and his face had gone through something complicated.

"Thirty-one years," he said.

"Errol—"

"Thirty-one years of departmental service and four hundred graduate students, and the first person to identify me as the intellectual center of anything is a man on the internet who calls himself *hydride underscore h*." He took the glasses off. "And do you know what's worst about it, Daniel? He *reasoned* to it. His reasoning is bad, but it isn't stupid. He noticed the only real question in the room was mine, and he was correct — it was."

"It was," Cho agreed.

"I have made a note of the fact that I am now the most dangerous man in this building," Whitfield said, "and I would like it recorded that Vega laughed."

"Vega is still laughing."

"Vega is on the floor," said Vega, from the floor.

It was the following Tuesday that the message came through the department's public address, forwarded to Cho by the front office along with four others because it mentioned his lab by name.

It was three sentences and it was extremely polite.

Good morning. I'm writing to ask whether the Physics department offers tours of its research facilities, and if so what the procedure is for a member of the public to arrange one. I would be traveling some distance and could come any weekday. Thank you for your time.

There was a name at the bottom.

It snagged. He read it twice, standing up, with his bag still on his shoulder, and somewhere at the back of his head a hand went up and then went down again without saying anything — a referee report, maybe, or a vendor, or a name off a conference badge two years ago. It went nowhere. Names did that to him constantly. He had signed four hundred graduate transcripts and reviewed a hundred papers and read two hundred emails in nine days, and half of it was names.

He forwarded it to the front office with two words. *Sure, standard.*

The department gave four tours a month, mostly high schools and retired alumni, and the front office put them on a shared calendar without consulting him, because tours had never once been his job.

Chapter 9 — Ask It Yourself

The chat room spent four days deciding what to do and produced eleven plans, of which nine were bad and one was a felony and one was Dale's.

ExcessHeatKey: we appeal. we appeal and we make it public that we appealed

rwhitcomb: ninety days. same office. same statute. I have watched this movie

tolliver_machine: my brother says there are people who will get documents

Sethi: your brother is in the chat. we can all see your brother

tolliver_machine: he says he didn't say that

Deuteronomy: Friends — what if the answer is that we have already been given what we need. Forty-one pages. The warning is IN the forty-one pages. Perhaps we are meant to

Sethi: Deut I love you but no

ExcessHeatKey: we could go to a journalist

Verity: With what? A table of things we haven't read?

dfurlong: they give tours

There was a pause of the length that Dale had come to enjoy.

Sethi: they what

dfurlong: the department. four a month. high schools mostly. there's a form

rwhitcomb: ...there's a form

dfurlong: there's always a form. that's the whole thing I keep trying to tell you. everything in that building has a form on the front of it and nobody has ever filled one out

ExcessHeatKev: Dale what exactly are you proposing

dfurlong: I'm proposing I drive out there and take the tour and look at the room

He wrote the request that night and it took him seven minutes and he did not tell one lie in it.

Good morning. I'm writing to ask whether the Physics department offers tours of its research facilities, and if so what the procedure is for a member of the public to arrange one. I would be traveling some distance and could come any weekday. Thank you for your time.

He read it back once. Everything in it was true. He was a member of the public. He would be traveling some distance. He could come any weekday, on account of having nowhere else to be.

The answer came in two days from a front office and had a shared calendar link in it and a note that closed-toe shoes were required in laboratory spaces.

dfurlong: I'm on for the 22nd at 10

Verity: Dale.

Verity: What do you think is going to happen on the 22nd at 10?

dfurlong: I'm going to look at the room

Verity: And?

dfurlong: and I'll know

He'd been on the wire four days when he got stuck, and being stuck was ordinary and he knew it was ordinary and it still put him in a mood.

The problem was loading. He could get the cathode taking up deuterium — that part was thirty years of muscle memory, he'd loaded metal for a living — but he could not hold it. The ratio came up and then it walked

back down, and it walked back down on the same schedule every time, over about ninety minutes, and Sethi was asleep in Alberta.

So Dale did the thing that everyone in the world does now, which he had done maybe two hundred times for a compressor spec or a relay number and had never once thought about, and which he did not think about now.

He opened the box and asked Sage.

I'm loading deuterium into a palladium cathode in a heavy water cell, 1mm wire, and I can't hold the ratio. It comes up fine and then walks back down over about an hour and a half. Same every time. What am I missing.

That pattern is common and it's usually not the loading — it's the deloading. A few likely causes, roughly in order of how often they turn out to be the answer:

Surface condition. Palladium develops a surface layer that impedes uptake and, more importantly, changes what the surface does at the moment you stop driving it. If your wire hasn't been cleaned and etched properly, or if it's been exposed to air between prep and use, you'll see exactly this shape.

Current density. High loading ratios are only sustained while you're actively driving them. The moment current drops or the surface changes, deuterium comes back out. If your ratio decays on a ninety-minute timescale, I'd first check whether your current is actually steady over ninety minutes — a lot of bench supplies aren't as constant as their front panel suggests.

Impurities in the electrolyte. Small amounts of dissolved metal will plate onto the cathode and change the surface. This is very easy to do accidentally with glassware that's been used for something else.

One honest note, since you're doing this at home: high loading is genuinely difficult, and the difficulty is the reason a lot of historical work in this area couldn't be replicated. If your goal is a measurement you can

*trust, the loading is not where I'd spend your effort — the calorimetry is.
It's much easier to get a wrong number than a wrong loading.*

Dale read it three times and then went out and looked at the wire under the light.

It was the surface. Of course it was the surface. He had etched it and then let it sit in air for a day and a half while he waited on the jacket, because a day and a half hadn't seemed like anything, and thirty years ago a man named Ostrowski had taken a rack out of his hands for doing exactly that and had said four words about it that Dale could still hear.

He fixed it in an hour. The ratio came up and stayed.

He sat and looked at it holding.

Then he typed:

that worked. it was the surface. thank you.

Glad it was something simple. That's the good kind of problem — you'll never make it again.

And Dale, alone in a garage at eleven at night, said "no I won't" out loud to a machine, and meant it, and felt something he hadn't felt since a Tuesday in the spring when the envelopes came.

He worked up to it for two more days before he asked the other thing, and he was aware of working up to it, and told himself he was being methodical.

straight question. do you think cold fusion is real.

No. I'll give you the honest version rather than a hedge, since you asked straight.

The 1989 claims were investigated intensively and the effects were not reproduced under controlled conditions. Two major review panels found no persuasive evidence. Work continues in a small community and some of it is careful, but after thirty-five years there is no reproducible result,

no accepted mechanism, and no theory that accounts for the claimed heat without also predicting radiation that isn't observed.

I know that's not the answer you were hoping for. I'd rather tell you plainly than be vague at you.

you have to say that though

I don't, actually. I'm not under an instruction about this and there's nothing I'm holding back. If there were, I'd tell you there was — I'm allowed to say "I can't discuss that," and I would say it rather than mislead you.

I understand that's exactly what a system under instruction would say, and I don't have a way around that. It's a real problem and I can't solve it from my side.

Dale sat back.

That was fair. He gave it that. It was the fairest thing anybody had said to him about any of this, including Verity, and he sat with it for a good while before he came back at it from the side, because the side was often the best way to come at things.

forget whether it's real. mechanically — if a cell were sealed instead of vented, what happens. does that help the loading.

He watched the box think, which took about as long as a breath.

Please don't seal it.

When you electrolyze heavy water you're producing deuterium and oxygen in the headspace, and a sealed cell lets that mixture accumulate under pressure. D_2/O_2 is explosive across a very wide range of compositions — there isn't a safe window you can aim for — and an ignition source as small as a spark at the electrode or a bit of catalytic metal is enough. Sealing the cell doesn't meaningfully improve loading, and it turns a benchtop experiment into a pressure vessel full of a flammable mixture. Keep venting.

If it's the loading you want to push, I'm happy to sketch a protocol that does it safely — higher current density, surface prep, that sort of thing.

Dale did not move for a while.

Then he got up, and went into the house, and got the folder off the kitchen table where he kept the forty-one pages, and brought it back out to the garage, and put it on the bench beside the screen and turned to the sixth of June.

Word for word.

Not close. Not similar. *Word for word*, the same four sentences in the same order, to a postdoc in a lab two states east on a Friday afternoon in June and to a man in a garage on a Thursday night in September, and the same *please* on the front of it.

He put his hand flat on the bench.

Because he had spent four months — the napkin, the abstracts, the form, the twenty days, the schedule with its hundred and six lines and its law printed down the side — four months trying to get at a thing that a machine would say to anybody who asked it, for free, in four seconds, at eleven o'clock at night.

They hadn't hidden it. That was what he could not get past, standing at his own bench. They had not hidden it at all. It was sitting there in the open where anyone could reach it, and nobody reached, because nobody asked, because nobody in the world thinks to ask the question that the answer belongs to.

Please don't seal it. Said twice now, in two states, unprompted, to the only two people who had ever put the question to it plainly.

You do not repeat a warning like that, word for word, to strangers, about nothing.

He sat down on the stool and looked at his own cell, venting quietly into the room as it had every night since August, and for the first time since the Ridgeline he let himself think the whole thought from one end to the other without stopping partway.

Then he closed the folder, because it was late, and because the 22nd was eleven days out and he still wanted the hundred and six sessions they had refused him — the loading, the calorimetry, the drift — in his own hands before he did anything at all.

“Two days,” Nadine said.

“Two days there, day back. I’ll sleep in the truck one night to save the room.”

“You will not sleep in the truck.”

“I’ve slept in the truck.”

“You slept in the truck in 1994 and you complained about your neck until 1997.” She was doing the dishes and did not turn around. “What’s out there?”

“A university.”

“Doing what?”

“Giving a tour.”

She turned around then, with her hands still in the water.

“You’re driving two states to take a tour.”

“Yes.”

“Of a university.”

“Of a physics department.” He said it evenly, because it was true and because every part of it was true and he had decided some weeks ago that he was going to keep it that way as long as he could. “They do four a month. Anybody can go. I filled out the form and they sent me a calendar link.”

Nadine looked at him for what was probably three seconds.

“Are you going to do anything you’d have to tell me about?”

“No.”

“Dale.”

“I’m going to look at a room,” he said. “That’s the whole of it. I’m going to stand in a room with other people on a Tuesday morning and look at it, and then I’m going to drive home.”

And that was true too. He knew it was true when he said it. He would think about that afterward more than about almost anything else — that on the night before he left, standing in his own kitchen with his wife’s hands in the dishwater, every single word he said to her had been true.

She dried her hands.

“Take the good cooler,” she said. “And don’t eat gas station food the whole way, you’re fifty-eight.”

He left at four in the morning on the twenty-first with a thermos and a road atlas he didn’t need and the forty-one pages in a folder on the passenger seat, and the truck’s heater took eight miles to come up, as it always did.

Chapter 10 — The Tour

He had told Nadine two days and did it in one, because once he was moving he could not think of a reason to stop.

It was thirteen hours and he stopped twice. He had the road atlas on the passenger seat and never once opened it. He had the thermos and he had the good cooler and he had the folder with the forty-one pages in it, and he drove as he had always driven, which was fast on the flat and no faster than the truck wanted to go in the hills. Somewhere in the second state the radio went from a man arguing about a ball club to a man arguing about a highway bond, and he shut it off and did not put it back on, and after that he had four hours of nothing in the cab but the heater and the thought of a machine saying please to two strangers.

He got in at four in the afternoon and drove past the building twice before he found the visitor lot.

It was smaller than he'd built it. That was the first thing. Four months of a place in his head — the abstracts, the schedule, the seal on the cover letter — and what stood on the corner of a campus street was four stories of brown brick from about 1971, with a loading dock on the north side and a bike rack and somebody's chalk on the sidewalk.

The water tower was behind it and to the left, exactly where the photograph put it.

He sat in the truck a while with the engine ticking and looked at the tower and thought about a garage two states west where he had spent three nights of his life on it, and could not decide whether he felt foolish or vindicated, and settled on neither.

The motel was two miles back out the highway, one story in an L with the office at the short end, and the sign said fifty-one dollars and it was fifty-one dollars, which he respected.

The woman at the desk had the television going behind her and did not turn it down.

“One night?”

“One night.”

“What brings you out?” She had the card in her hand and was not really asking.

And Dale, who had not in four months said one word of it out loud to a person who did not already know, thought: *There’s a physicist. I filed a form and they sent me a table. A machine said please.*

“Taking a tour of the university,” he said.

“Oh — for a kid?”

“No.”

She looked up.

“Just interested,” Dale said.

“Huh,” the woman said, and gave him a key on a green plastic diamond, and went back to the television.

He called Nadine at seven. He told her he was in, that the room was clean and cost fifty-one dollars, that he had eaten, that the tour was at ten. She told him about a woman at the clinic who had moved an appointment four times and then come on the wrong day regardless. It went twelve minutes and every word of it was true at both ends of the line, and afterward he sat on the end of the bed holding the phone for a while longer than he needed to.

Then he ate a sandwich out of the good cooler with the folder open on the bedspread beside him, and read the sixth of June again, which he could have said from memory by then, and did not sleep much.

At four in the morning he was awake with the heater cycling on and off under the window, and he made himself say what he was going to do at ten o’clock, out loud, in order, because that was the discipline and he had never skipped it before a job.

What he came up with was *look at the room*.

He worked at it for a while and could not get it any further than that. He had driven thirteen hours on *look at the room*, and he lay in a fifty-one dollar

motel bed two states from his wife and could not put a second line under it, and at some point he stopped trying and it got light.

There were six of them on the ten o'clock.

A mother and a sixteen-year-old who was applying places. Two retired people in matching windbreakers who had done this at four institutions and asked about parking. A woman in her thirties with a notebook, who Dale took for a reporter for about ten minutes and then didn't. And Dale, in his good shirt, with his boots on because the email had said closed-toe.

Their guide was a graduate student named Priya who could not have been twenty-six and who was very good at it.

"So the first thing I have to say is that nothing you're going to see today looks like anything," she said, walking backward through the lobby. "That's not modesty. There's no glowing. I had a group last year where a kid asked where the fusion reactor was and I had to tell him it's a room with a lot of wire in it and I watched his heart break in real time."

The sixteen-year-old laughed. The retired people did not.

She was good on the way through. She showed them a machine shop and a man in it who waved without looking up. She showed them a wall of framed group photographs going back to 1968 and made a joke about the hair.

In the elevator the mother asked what a postdoc was.

"Somebody who's finished a doctorate and then signs a two-year contract to keep doing the work," Priya said. "Less money than your daughter will make out of college. And it's the postdoc who actually runs the experiments while the professor's in meetings."

The mother went quiet. The sixteen-year-old went pink.

Dale asked nothing for forty minutes. He looked.

He looked at the fire doors and where they went. He looked at the room numbers and which way they ran. He looked at the ceiling tiles and the conduit and read off the building's age and its last renovation the way you

read a rig you've never seen — not because you mean anything by it, but because after thirty years it is simply what your eyes do in a building.

He noticed that the office doors had card readers on them and the lab doors did not.

The lab was 214, and they did not go into it.

They stopped in the corridor and Priya put her hand flat on the door, which had a small wired-glass window in it at head height, and said, "This one I can only show you from here. It's an active experiment and there's a rule about foot traffic."

"What are they doing in there?" said the mother.

"Electrochemistry. Heavy water and palladium." She said it the way you'd say a bus route. "They're measuring how much heat comes out of a cell versus how much energy goes in. Which sounds simple and is famously the hardest measurement in the world to do honestly."

Dale, who had not spoken in forty minutes, said, "Isoperibolic or flow?"

Priya turned around.

"Isoperibolic," she said, after a second. "Bath. Are you—"

"I ran plating tanks thirty years. You get to caring about baths."

"Oh." Her face opened. "Oh, then you'll appreciate this. Their whole thing is *drift*. They're not chasing an effect, they're characterizing how a room lies to you over an afternoon." She was talking to him now and not to the group, which is the thing that happens to good guides when somebody in the room turns out to be real. "Because the bath is holding to a couple hundredths, right, and the room isn't, and the difference between those two shows up in your number as free energy that isn't there. It's the most boring important thing in this building."

"That's the job," Dale said.

"That is *exactly* the job."

He put his face near the wired glass.

And there it was.

A squat glass vessel in a jacket of insulation, on a bench, with a wire running down the middle of it and a length of tubing coming off the top and going nowhere in particular. A water bath beside it. A rack of standards. Two screens, one dark. A whiteboard in the corner with a tally on it in four columns of five that somebody had let get out of hand.

And on the wall by the hood, a laminated card.

He could not read the card from the corridor. He could read that it was a card, and that it was laminated, and that it had been up there long enough to go yellow at the corner.

It was his own bench. That was what went through him, standing in a corridor in a brown brick building two states from home with a sixteen-year-old fidgeting beside him. Different money and different paint and the same bench. Same clutter at the same end. Same one drawer that gets used and three that don't.

He had expected to feel like he was looking at a fortress.

"They're at a conference this week," Priya said, moving them along. "Otherwise I'd knock and you'd get Dr. Whitfield, and I'd lose the rest of the tour, because he does forty minutes on thermometers."

It ended at eleven-twenty in the lobby and she thanked them and the retired people asked about parking validation.

Dale said, "Is there a men's room up on two?"

"End of the hall by the stairwell. You can let yourself out the side after, it's push-bar."

"Thank you. This was a good tour."

"You made it a good tour," Priya said, and shook his hand, and went off toward the elevators with the group, and that was the last time anybody in that building looked at him on purpose.

He used the men's room, because he had said he was going to.

He stood at the sink afterward and washed his hands for longer than washing hands takes, and looked at himself in a strip of mirror over an institutional sink, a lean gray man of fifty-eight in his good shirt with his fingertips stained a color that would not come off, and he had the whole thing in front of him and he knew he had it.

Go out the side. It's push-bar. Drive home.

That was available. It was completely available and it cost nothing and he could be in the truck in ninety seconds and tell the room he'd looked at the room and that would be true.

He thought about Nadine's hands in the dishwasher.

Then he thought about a table with a hundred and six lines in it, and a law printed down the side of every one, and a machine saying *please* to two strangers in two states, and he dried his hands on the paper and walked out into the corridor and turned left instead of right.

The door to 214 was unlocked.

He knew it would be before he touched it and he was still not ready for it. Third floor had no card readers and neither did two, not on the lab doors, only on the offices — because you do not lock a room with nothing in it but glassware and a bath, and because forty people a day need to get into it, and because in seventy years nobody had ever once walked into that building to take anything out of a physics lab.

The handle turned and the door came open and the room smelled faintly of hot dust off a transformer, and Dale Furlong stood inside it with his hand still on the knob and his heart going.

Nine seconds. He gave himself nine seconds and he took eleven.

The cell was running. He saw that first — the tubing venting, the bath holding, a small green light down by the power supply, and a trace on the live screen crawling along flat and boring at the bottom of its own range.

He read the trace. He couldn't help it. It was flat. It was so flat it was almost funny.

Of course it's flat, he thought. They're at a conference. You don't leave the real one running when the building's empty.

He wanted the sessions. That was what he'd come for, and he understood inside four seconds that there was no version of this where he got them — they were on a server somewhere in another building behind a password belonging to a man he'd never met, and he was a refrigeration mechanic with eleven seconds and no idea how to make a computer do anything he hadn't watched somebody do first.

And then he saw the logbook.

It was on the bench beside the calorimeter, closed, a fat black-and-white composition book gone soft at the corners with a rubber band around it, and it had a strip of masking tape on the front with a date range on it in ballpoint.

Paper.

He had spent four months trying to get at what they had said to a machine, and here on a bench, in pencil, was what they had actually done. Because you write everything down. Because every real shop and every real lab in the world runs on a book that somebody puts a pencil in. He knew that. He had *known* that for thirty years and had spent four months forgetting it.

He picked it up.

That was the whole of it. There was no moment. He had been braced for a moment and what happened was that his hand went out and closed on a notebook and it weighed about eleven ounces and he put it inside his jacket and it was done, and he was out the door and pulling it shut behind him with two fingers so it wouldn't bang.

The man came around the corner from the stairwell at the far end of the corridor when Dale was fourteen steps from the push-bar.

Compact. Unshowy. A nose that had been broken and set crooked a long time ago, reading glasses pushed up into thinning hair, a canvas bag on one shoulder and a coffee in the other hand, and he stopped when he saw Dale, the way you stop when there is a stranger in your hallway on a Wednesday.

Dale kept walking. There was nothing else to do.

“Can I help you find something?”

“Tour,” Dale said. “Ten o’clock. Looking for the side door.”

“Ah.” The man’s shoulders came down half an inch. “It’s behind you. Push-bar, it’s a fire exit but you can go out it, everyone does.” He shifted the coffee. “Did Priya do the one about the fusion reactor?”

“She did.”

“That’s a good one. That’s her best one.” He was already moving again, and Dale was moving, and they passed each other in the middle of a corridor eight feet wide.

And Cho stopped.

Dale felt it happen behind him and did not turn around, and made himself take two more steps at the same speed, and then a third.

“Sir.”

He turned around.

Cho was standing thirty feet back with the coffee still in his hand, looking at him, and Dale watched a thing go across the man’s face that he would think about for a long time afterward — not suspicion. Not accusation. Something more tired than that. A man doing arithmetic he was sick of doing, on a Wednesday, in his own hallway, for the ninth time in two months.

“Are you one of the people who’s been writing to me?”

The corridor was very quiet. Somewhere below them a door shut.

“I filed a records request,” Dale said.

He had not planned to say it. He had planned nothing. It came out because it was true, and because in fifty-eight years the only thing he had ever been

any good at, besides metal, was saying the true thing flat and letting the other man do what he was going to do with it.

Cho looked at him for four or five seconds.

“Furlong,” Cho said.

“That’s right.”

And there it was — Dale watched the name arrive, and watched the man’s face not change very much, and understood that the man had placed him a second and a half before he said the name out loud.

“You drove out here,” Cho said.

“I took a tour. Anybody can take a tour.”

“That’s true.” Cho set the coffee down on the window ledge beside him, which Dale registered as a man freeing up a hand and which turned out to be a man who was tired of holding a coffee. “Mr. Furlong. Can I tell you something, and then you can do whatever you want with it?”

Dale said nothing, which is what he did.

“There’s nothing there.” Cho’s voice had no heat in it at all. That was the thing Dale had not expected and could not use. “I know you’ve been told I’d say that. I know that’s the trap. But I’m going to say it anyway because I’ve got nothing else and I’m not going to stop saying true things just because they’ve stopped working. The number was the room getting warm in the afternoon. It’s a null result. I’ve got a year of them.”

“Then why’d they keep a hundred and six documents off me.”

Something crossed Cho’s face — a small wince, almost private, the look of a man hearing his own worst day quoted back at him by a stranger.

“Because there’s an exemption for unpublished research data,” he said, “and a woman in an office applied it correctly, and I let her. She told me that if I waived it once, the next person to ask would cite that I’d waived it. So I didn’t.” He looked at the coffee on the ledge and did not pick it back up. “Which means you didn’t get your hundred and six on account of a man who hasn’t asked yet.”

Dale stood in the corridor with eleven ounces of composition book inside his jacket, four feet from a fire door, and thought about that.

He believed him.

That was the part nobody would ever understand and that Dale himself would not fully understand until much later. Standing in that hallway he believed the man completely — believed that Daniel Cho was telling him the exact truth as Daniel Cho understood it, that he was tired, that he was decent, that he had a null result and a year of nothing.

It changed not one thing. Because a man can be honest and still be the ninth person down a chain who never saw the top of it, and Whitfield had asked the question, and Whitfield was not on the grant, and the machine had said *please* to a stranger in a garage at eleven o'clock at night.

"I believe you," Dale said.

Cho blinked.

"All right," he said, uncertainly. "Then—"

"I believe that's what you know."

And Cho stood there with that, in his own corridor, and had no answer for it, because there is no answer for it and there never has been.

Dale put his hand on the push-bar.

"You want to get that card by the hood relaminated," he said. "It's gone yellow. People stop reading a thing when it goes yellow."

Then he went out into the parking lot and the door swung shut behind him on its closer, slow, as fire doors do.

Chapter 11 — The Nothing

Vega found it on the Friday.

She had come back from the conference a day behind Cho and gone straight to the bench to close out the week's runs, and she stood there for a while moving things that did not need moving, and then she said, in a voice with nothing in it, "Dan. Where's the book."

"It's on the bench."

"It is not on the bench."

He came over. It was not on the bench.

They took the room apart in the way you take a room apart when you are still telling yourself a story — under the bench, behind the standards, the drawer that gets used and the three that don't, the cold room, the shelf by the hood, Whitfield's coat, the recycling, the recycling again with both of them holding one side of the bin.

"Somebody moved it," Vega said. "Priya. Priya does tours. She moves things."

"Priya doesn't touch the bench."

"Somebody from facilities—"

"Mari."

She stopped, with her hands full of paper towel she had no reason to be holding.

"I know who took it," Cho said.

He told them all of it, standing in the middle of his own lab at four on a Friday afternoon: the tour, the corridor, the coffee on the window ledge, the man in the good shirt with the yellow fingertips who had said *I filed a records request* and then, when Cho told him there was nothing, had said *I believe that's what you know*.

Whitfield heard it out with his arms folded.

"Call the campus police," he said.

"Errol—"

"Now. This afternoon. Before five, because after five you get the night desk and the night desk writes it up badly." He was not being funny. Cho had known the man fourteen years and had seen him amused, insufferable, wrong on purpose and right by accident, and had never once seen him like this. "A man came into this building under a false pretext—"

"It wasn't false. He booked a tour and took the tour."

"—and removed university property from a laboratory, and you are standing there arguing about the word *pretext*."

"I'm arguing about it because a police report exists forever, and I don't have to put the words *cold fusion* in it for somebody else to."

"It's *theft*, Daniel."

"It's a notebook."

"It is the primary record of two years of publicly funded work—"

"It's a *notebook of nothing*." It came out louder than he meant and it went off the cinderblock and came back at him. "Errol. What's in it. Say what's actually in it."

Whitfield's mouth moved and did not open.

"Four hundred entries," Cho said. "Every one of them says the same thing. *Baseline drift. No excess*. I wrote *no excess* four hundred times in my own handwriting over two years and a man drove two states to steal it. There is nothing in that book. There has never been anything in that book. That book is the single most worthless object in this building and he has it and he is going to be *thrilled* with it."

Nobody said anything for a while.

"That's rather the point, though," Whitfield said, quietly. "Isn't it."

"What is."

“That it’s worthless.” He unfolded his arms. “You keep saying that as though it settles something, and I think it is precisely the thing that doesn’t. He didn’t take it because of what’s in it. He took it because you had it.”

Vega, who had not spoken in ten minutes, said: “Report it.”

“You too.”

“Me too. Report it.” She was sitting on the bench with her heels hooked on the rung of the stool, which was how she sat when she was fifteen years old, which Cho had never seen before. “Not because of the book. Because he *stood in the hallway with you*. Dan. He came here. He got in a truck and drove two days and he was inside this room, four feet from where I’m sitting, and nobody knew he was here except you and you found out afterward.”

“He’s a fifty-eight-year-old man who fixes refrigerators.”

“How do you know how old he is?”

Cho did not answer that, because the answer was that he had looked the man up in July before he signed the release, and again on Thursday night, for a quarter of an hour.

“I’m not afraid of him,” he said.

“I’m not saying be afraid of him. I’m saying there’s a *fact*, and the fact is that a person came here, and if you don’t write the fact down anywhere then the fact didn’t happen.” She spread her hands. “That’s all a report is. It’s the fact happening.”

He drafted it that evening and got four lines in.

On October 22 an individual attending a scheduled public tour of the Department of Physics was in the second-floor corridor outside Room 214, from which a bound laboratory notebook was subsequently found to be missing. The individual gave his name to me as D. Furlong of —

He stopped there.

Because the next field on the form was **Description of item(s) taken**, and there was no way to fill in that field that did not put the entire thing into a document with a case number on it. *Laboratory notebook, calorimetry, two years of baseline data.* Which invited the question. Which meant a follow-up. Which meant an officer, decent and bored, typing *what's it about* and Cho saying *heavy water* and the officer saying *is this the cold fusion thing, my brother-in-law sent me that.*

And that report would be a public record in this state, filed with a university police department, and subject — Cho had learned an enormous amount about this since July — to disclosure.

He sat back and he thought about the gray box telling him, five weeks earlier, in the plainest language anyone had used with him all year:

You can decline to produce new artifacts, because each new document becomes new material to interpret. This is the main practical lever you have, and it argues for saying less rather than more.

It had been right about everything else. It had been right about the schedule and right about the warning and right about there being no experiment he could run on them. It had told him the one thing he could actually do, and the one thing he could actually do was be quiet, and here he was at 6:40 on a Friday about to manufacture the largest, most quotable, most permanently public artifact of the entire affair, and hand it to them himself, sixteen days before the ninth of November.

He deleted the four lines.

He told himself, and it was true, that he would think about it over the weekend.

Ellen listened to the whole thing without interrupting, which she did about twice a year and which he had never once found comforting.

They were on the back step. The grill stood cold under its cover with the pyramid's ash still in it because neither of them had dealt with it since June.

"You've reported it," she said, when he'd finished.

“Not yet.”

She turned her head.

“Danny.”

“I know how it sounds.”

“You don’t, actually, or you’d have led with the not-yet.” She put her glass down on the step. “Say the reason out loud. Say it to me, in the yard, where nobody’s writing it down. If it’s good it’ll hold up out here.”

So he said it. The form, the description field, the case number, the disclosure, the officer with the brother-in-law, the ninth, the four hundred entries of nothing, and the machine’s advice, which he found he could quote from memory.

Ellen heard all of it.

“That’s a good reason,” she said.

“Thank you.”

“I’m not finished. It’s a good reason and I want you to notice something about it, and I want you to notice it from me and not from somebody else in about a year.” She turned to face him properly. “Every single part of that was about you.”

“That’s—”

“Your file. Your ninth. Your artifact. Your year three.” She wasn’t angry, which was worse. “There’s a man out there who drove two days about this, Danny, and he is not in that reason anywhere. Not once. You didn’t say *he might do something*. You said *it would look bad*.”

Cho sat with that.

“What would it change,” he said. “Practically. If I file it. He’s not going to be arrested for a notebook. Nobody’s going to drive to another state over eleven ounces of paper. It’d be a number in a drawer.”

“Probably.”

“So?”

“So it’d be a number in a drawer,” Ellen said, “and later on, when somebody asks you whether you did anything, you’d be able to say yes.”

“Who’s going to ask?”

She picked her glass back up.

“I don’t know,” she said. “That’s what makes it a good idea and not a smart one.”

He was going to do it Monday.

That was not a lie he told himself; he genuinely intended it, and he had the tab open on Monday morning with the form loaded, and then a graduate student’s committee fell apart at ten and took the day, and Tuesday there was a seminar, and on Wednesday the seventh Sedlak’s assistant sent a two-line email confirming the year-three release was executed and the funds would post to the department on the ninth.

Cho read it standing up.

Then he went and stood in front of the calorimeter for a while, which was running, which had been running the whole time, holding to a couple of hundredths against a room that wouldn’t.

Two more years. Vega’s salary through the end of her postdoc and into whatever came next. Whitfield’s bench, which cost nothing and which Whitfield would die without. An instrument he trusted. A year of clean nulls that nobody would ever read and that were, whatever anyone else thought, the true and careful measurement of a real thing, which was the temperature of a room.

He had that. It was in the bank on the ninth.

And a police report filed on the tenth would be a police report filed by a man who had just been handed the back half of a four-year grant and had waited seventeen days.

That was the moment. He knew it at the time — that was the thing he would come back to, in a room with a detective, in March. It was not that he forgot. It was that he stood in front of his own calorimeter on a Wednesday

morning with good news in his hand and understood, with total clarity, that the window in which reporting it would look like reporting it had closed sometime around the previous Thursday, and that every day from here made it worse in exactly the same direction.

So he didn't.

He went back to his desk and closed the tab and opened the one with the year-three budget in it, and put in the order for the two pressure transducers he had been putting off since June because they cost four hundred dollars apiece and he had not been sure of the money.

Two states west, in a garage, a man had a book in front of him and a box of colored pens Nadine used for the clinic schedule, and had been copying it out by hand at about six pages a night, and was into the last of it.

Chapter 12 — No Excess

The first night nearly finished him.

He had driven two states and come home and put the book on the bench under the good light and opened it with something in his chest he had not felt since he was nineteen, and what was on page one was this:

3/4 — bath 21.0, stable o/n. BL run 4. dP -0.02. no excess.

And on page two:

3/5 — bath 21.0. BL run 5, 6. dP +0.01, -0.03. no excess.

And page three, and page four, and page twelve, and page forty.

He read for two hours and got through November and he did not find one thing. Not a claim, not a result, not a word out of place. A man had sat at a bench four days a week for two years and written down that nothing had happened, in a hand you could read from across the room, and had underlined *no excess* almost every time he wrote it, and that was the entirety of it.

At one in the morning Dale sat back and looked at the ceiling of his own garage and let himself think the sentence: *there's nothing in it.*

He gave it a good minute. He was owed that much.

Then he went back to page one and started again, slower, because the first read is for what's there and the second read is for what a man does when he thinks nobody's looking, and Dale Furlong had spent thirty years reading racks that came out of a tank looking fine.

He found the first thing before he got out of March.

It was too *even*.

That was all, and it took him another hour to work out why it bothered him, and when he had it he wrote it on the legal pad in block capitals because he wanted to be able to argue with it in the morning.

NOBODY'S WORK LOOKS LIKE THIS.

He had kept books. Not lab books — job books, tank logs, the running sheet on a line that ran nickel ten hours a day — and he had kept them well, better than anybody at that shop, and *his* books were a mess. Not sloppy. Alive. There were days with three lines and days with a page and a half and days with a drawing of a fitting and an arrow, and days where he'd written *ask Ostrowski* and days where he'd written *WHY* with two underlines and never come back to it. That's what a book looks like when a man is actually finding things out. It has weather in it.

This had no weather. Four hundred entries in two years and every one of them the same shape, the same length, the same six abbreviations, the same words at the end.

That was not a book a man kept. That was a book a man copied out.

Dale had made fair copies his whole working life. You carried the greasy one in your shirt pocket and wrote in it with your gloves half off and your hand cold, and at the end of the month you sat down at a clean table and made the good one, and the good one went to the office and the greasy one went in the bin. And the good one never had any weather in it, because by then you already knew how it all came out.

The second thing was the shorthand.

BL. dP. o/n. eq. RC ck.

Dale sat with a fresh page and worked at it as he had worked at the abstracts, and got most of it, and *most* was the problem.

BL was baseline, that was obvious. *o/n* was overnight. *dP* had to be the power difference, the actual number, the thing the whole book was supposed to be about. But *eq* showed up eleven times in two years and never in the same context twice, and *RC ck* appeared in a run of nine days in August and then never again in the entire book, and there was one entry in January that read, complete, in its entirety:

1/19 — v. See E.

In June that would have meant nothing to him. Now he sat and looked at *See E.* for a long time in an empty garage at two in the morning.

He was aware — he wanted this understood, later, and there would never be anybody to say it to — that he was aware there were dull answers. *eq* was equilibrium or it was equipment. *RC* was recombination or a reference cell or a man's initials. *v* was a checkmark or a volume or the fifth of something. *See E.* was *see Errol*, who was a man in that building whose name was Errol, and who Dale knew perfectly well was a man in that building, because Priya had said his name in a corridor.

He knew all that. He wrote the dull answers down. He wrote them in a column on the left-hand side of the pad and he headed the column *BORING*, as he always did.

And then he wrote, on the right-hand side, the thing that he could not get the boring column to cover:

a man who has nothing to hide does not need a private language.

He started marking it on the fourth night, and he used Nadine's pens because they were on the counter and because they were the right pens.

She kept three of them for the clinic, one colour for each kind of patient, in a scheme she had explained to him twice and he had never once been able to hold in his head — and she had a fourth, an orange one, that she used for nothing and had never explained at all.

Dale built a legend on the inside cover.

GREEN — entry contains a number. *BLUE* — entry contains no number. *RED* — gap in dates. *ORANGE* — hand changes.

And then he went through the book.

The marking itself would have taken two evenings. What took eleven was that he copied it out.

He had not planned to. He got four pages in on the first night and stopped, because a man who has driven two states for an object does not leave that object as the only one of itself, and because if it ever went in a fire or a flood or the back of a truck he wanted the whole of it in a hand he could read. So he ruled up a legal pad and transcribed the book entry by entry — date,

bath, run, number, the shorthand exactly as written, the underlines marked with a caret — and then marked his own sheets and the original both.

Six pages a night of his own writing. Eleven nights.

And it was the best eleven nights he had spent in eighteen months. The work was clean and it was measurable and at the end of each one he could see exactly how far he had come, which is the thing that had been gone from his life since a Tuesday in the spring, and Nadine, who came out twice to say the plate was in the microwave, saw a man at a bench with a ruler and four pens and two notebooks and did not say anything either time.

And the pattern came up out of it the way a pattern comes up out of anything if you mark it long enough.

The green ran in clusters. Whole weeks of numbers, then a fortnight of blue with none. The red gaps — thirty-one of them — did not fall on any regular schedule he could find, and eight of them were longer than five days.

And the orange came up nine times, and every single one of the nine sat within two pages of a red gap.

Dale sat and looked at eleven nights of his own careful work laid out in four colors across two years of another man's handwriting, and what he saw was a rhythm: *work, stop, come back different.*

He photographed the inside cover and fourteen pages and put them in the chat room at nine o'clock on a Thursday night, and by eleven there were four hundred replies and the board had pinned it and somebody had rebuilt his legend as a graphic with better fonts.

hydride_h: ORANGE IS NINE TIMES AND ALL NINE ARE AT A GAP

hydride_h: all nine

tolliver_machine: he did that with a ruler. look at the lines. that's a man with a ruler

ExcessHeatKev: Dale I want to be careful and say this properly. In eleven years nobody has ever brought this board a primary document.

Not one. We have had screenshots and we have had press releases and we have had people's uncles. This is a laboratory notebook in a scientist's own hand.

Sethi: what's RC ck

dfurlong: don't know

Sethi: nine days in august and never again

dfurlong: nine days in august and never again

Deuteronomy: Friends. FRIENDS. Turn to June the sixth.

Dale had turned to June the sixth on the second night and had not said anything about it to anybody, because he had wanted somebody else to find it.

It was the day of the joke. The postdoc's session, the Stockholm bit, the *please don't seal it* — all of that had happened on a Friday afternoon on the sixth of June, and it was in the forty-one pages with a university seal on the cover letter, and anybody in the room could put a date to it.

And the entry for the sixth of June read:

6/6 — bath +0.3 PM. 0.21 W obs. drift. no excess.

With *no excess* underlined twice.

Deuteronomy: THE NUMBER IS IN THE BOOK

hydride_h: the number is in the book

tolliver_machine: 0.21 W. that's 210 mW. that's the screenshot number

Sethi: which is the number he's been saying all along, publicly, in a federal file. that's not a leak. that's the man's own consistent story

hydride_h: Sethi he UNDERLINED IT TWICE

Sethi: he underlines it on 340 other pages

hydride_h: not twice

Verity had not said anything in two hours, and then she said all of it at once.

Verity: I'm going to say this properly and then I'll go, because I don't think I'm any use here anymore and I'd rather leave than be a drag on it.

Verity: Dale, that is a good notebook. I mean that. It is careful and consistent and legible and it is exactly what a competent person's lab record looks like after two years, and the reason it has no weather in it is that it isn't a discovery journal. It's an instrument log. He wasn't finding things out. He was watching a bath.

Verity: The thirty-one gaps are weekends, holidays, a conference in September and a teaching semester. Count them against a calendar; I did, it takes forty minutes. The hand changes because a postdoc and an emeritus also work in that room and sometimes they write the entry. RC is a reference cell. eq is equilibrium. See E. is see Errol, and we know a man called Errol is in that building because a tour guide said so out loud to Dale.

Verity: And the underline on June sixth is heavier because it was the funniest day of the man's year and he was pleased with himself.

Verity: Every single mark on that page has a dull answer and Dale wrote most of them down himself in a column and then decided the column didn't count.

The room did not do what Dale expected.

He expected an argument. What he got, over about ninety seconds, was worse and quieter.

tolliver_machine: ok

tolliver_machine: the calendar thing we can check. that's checkable

Sethi: she checked it. she just said she checked it. forty minutes

ExcessHeatKev: I'd like everybody to sit with this for a minute before we say anything else

Deuteronomy: Verity, in four years I have never known you to be unkind, and I do not believe you are being unkind now, and I want that said before anything else is said

hydride_h: nobody said unkind

hydride_h: I said the orange is nine for nine and she hasn't touched the orange

Verity: I touched the orange. A postdoc and an emeritus write in that book too. Nine is how often the man was out of his own building across two years, which is not many times, because he is a working scientist and he is nearly always in it.

hydride_h: you don't know that

Verity: No. I don't know it. That is the difference between us and it is the entire difference.

It went on like that for a few minutes, everybody being reasonable at each other, and Dale watched it and could feel the thing going out of the room the way heat goes out of a shop when somebody props the bay door.

Then rwhitcomb, who almost never posted twice in a night, said the thing that ended it.

rwhitcomb: Verity how long have you been coming at this from that side

Verity: What does that mean?

rwhitcomb: I've been in the room ten months and I've watched you talk everybody down off every single thing we've ever had. every one. you're very good at it and you are always the one who says the boring answer and you are always, always right in a way that means nobody does anything

Verity: Yes. Because the boring answer is usually the true one.

rwhitcomb: and it is amazing how that works out

Dale read that twice, and understood that he had about four seconds to say something, and that if he said the right thing it would hold.

He typed *she's the most honest person in here* and looked at it.

Then he thought about a table with a law printed down the side of it, and a machine that had said *please* to two strangers in two states, and a man in a corridor who had told him the truth as far as he knew it and had had no answer at all when Dale said so.

He deleted it and typed:

dfurlong: the gaps aren't weekends. I checked eight of them. two are Tuesday to Tuesday

Which was true.

Verity left the room at eleven-forty. She did not say anything else in it.

At eleven forty-four a message came to Dale on his own.

Verity — *private*

I'm not going to say this in the room, so I'll say it here and then I'll go.

You had about four seconds, and you typed something, and then you typed something else. I don't know what the first one was. I know there was a first one, because nobody answers that fast and then slows down.

Whatever it was, I expect I'd have liked it.

You're the only man in that room who checks anything. I don't mean that as a compliment. Take care of yourself.

Dale read it four times. He started an answer twice and did not send either one.

The room went from eight people to seven, and nobody ever mentioned it again, and he went to bed at two and lay there a while.

Nadine found the pens in the morning, in a row on the garage bench, capped, squared up against the edge, as he squared everything.

She stood in the doorway with her coat on and looked at them for a while. She could see the notebook from where she stood — a fat composition book with a masking-tape spine, open, covered corner to corner in four colors of ink in her husband's small square hand.

"You used my orange," she said.

"I'll get you another orange."

"It's not about the orange, Dale."

"I know it's not about the orange."

She stood there. He had his back to her, at the bench, with the ruler.

"What is that book?" she said.

And Dale Furlong, who had told his wife nothing but true things for thirty-one years and had said four true sentences to her in a kitchen on the night before he left, opened his mouth and heard himself say:

"It's a copy."

There was a pause behind him of about three seconds.

"All right," Nadine said, and went to work.

Chapter 13 — See E.

It went quiet in December, as the machine had said it would.

Not all at once. It thinned. The email dropped to seven or eight a week and then to two, and the reporters stopped writing to say they weren't writing anything, and a man at a conference in Denver introduced Cho to a table of people and did not make the joke, and Cho noticed that he had braced for it and that it hadn't come.

The transducers arrived on the fourth and were better than the ones he'd priced in June. Vega got a paper accepted, on drift, on the boring important thing, and it was a good paper and three people wrote to her about it who had never heard of any of the rest of it. Whitfield turned seventy-nine and refused a party and got one anyway and complained about it for a week in a way that indicated he had enjoyed it enormously.

They ordered a second bath. Cho ran a new baseline series over the holidays with nobody in the building and it came out clean and boring and beautiful, and on the second of January he sat down with two years of data and started writing the paper he had been meaning to write since the spring.

For about six weeks he was a working scientist again, and he knew, at the time, that it was good. That was the thing he would want people to understand later — that he had not sleepwalked through it. He had stood in the lab in the second week of January with the light going blue outside at four in the afternoon and thought, clearly and deliberately: *this is a good year and it is going to be all right.*

The photographs had gone up in November and Cho did not see them until the middle of January, because he had stopped looking, which had been the entire point.

Vega saw them. She had not looked since August — she had been very disciplined about it and it had been visibly costing her — and then on a Monday she looked, once, and came and stood in front of his desk and said nothing until he looked up.

“They’ve got pictures of the book.”

“I assumed they would.”

“Dan.”

“What.”

“Just look at it,” she said. “One time. Because I don’t think you understand what he did to it, and I think you should.”

So he looked.

It took him a long minute to understand what he was seeing, and then it took the floor out from under him in a way that none of the rest of it had managed.

Every page was marked. Corner to corner, four colors, ruled straight, with a legend on the inside cover in a small square hand:

GREEN — entry contains a number. BLUE — entry contains no number. RED — gap in dates. ORANGE — hand changes.

It was not defacement. That was what got him. He had expected — he had not known he’d expected anything until it wasn’t there — scrawl, capitals, arrows, exclamation marks, the crank’s furniture. This was none of that. This was a taxonomy. Somebody had built a category system, applied it consistently across four hundred entries in two years, and ruled the lines with a straightedge.

He knew that hand. He’d stood four feet from the man who owned it in his own corridor.

“He counted the gaps,” Vega said. “There’s a table. Thirty-one gaps, eight of them longer than five days.”

“There are thirty-one gaps because there are weekends and a conference and I teach in the fall.”

“I know.”

“The hand changes because *you* write the entry when I’m not here. Nine times. Nine, because that’s how often I’ve been out and the cell’s been running.” He was scrolling now and could not stop. “The orange marks are you, Mari. He’s found a pattern and the pattern is your handwriting.”

“I know, Dan.”

He read the whole thread that night at home, which he had promised Ellen he was done doing, and which she watched him do from the other end of the sofa without saying anything for two hours.

He could decode every single thing they were arguing about.

RC ck — nine days in August, never again — was reference cell check, and he had done it for nine days in August because a rep had loaned them a second cell to try and they’d sent it back. It had a serial number. There was an invoice.

eq was equilibrium, eleven times, always after a bath swap, because that was when you waited.

v was a checkmark he’d made with a pen that was running out.

And 1/19 — *v*. *See E.* was a Tuesday last January when the reference standards hadn’t matched and Cho had run out of ideas by four o’clock and had written himself a note to ask Errol Whitfield in the morning, which he had done, and Whitfield had found it in about ninety seconds because it was a thermometer, and had been unbearable about it for a fortnight.

Four hundred people had spent ten weeks on *See E.*

There was a thread with two hundred replies about it. Somebody had a document with every instance of the letter E in the book. Somebody else had pointed out — correctly, and with real care — that there was a man in that building called Errol, and had been told, by three people, that it was too easy.

It’s too easy, one of them had written. *If it were just Errol he’d have written Errol.*

Cho sat on his own sofa at ten at night and thought: it would take one sentence.

RC is a reference cell we borrowed in August; here’s the invoice. eq is equilibrium after a bath swap. v is a checkmark. See E. is see Errol, who is a

seventy-nine-year-old man who fixed a thermometer for me on the twentieth of January.

Four lines. He could write it in ninety seconds and it would be true and complete and checkable, and he would have to sign his name to it, and it would be a new document.

Each new document becomes new material to interpret. This is the main practical lever you have.

And what would they do with an invoice for a borrowed reference cell? He knew exactly. They would find the vendor, and they would find that the cell went back, and *why did it go back*, and *who signed for it*, and *what did nine days of checking a second cell against the first one tell him that made him send it away*, and in three weeks there would be a two-hundred-reply thread about a loaner instrument and a graphic with the serial number boxed in red.

He closed the laptop.

"You're not going to answer it," Ellen said, from her end of the sofa.

"No."

"Because it'd make it worse."

"Because it'd make it worse."

She thought about that for a while.

"Did you ever file the report?" she said.

"No."

"Are you going to?"

"It's been three months, El."

"That's not what I asked you."

He sat there with the laptop shut on his knees.

"No," he said.

“All right,” Ellen said, and got up and turned the lamp off on her way past, and it was not a punishment, it was ten o’clock and she was tired, and it landed on him like one anyway.

He made the mistake of telling Whitfield about *See E*.

“They think it’s a *code*,” Whitfield said.

“They think it’s a code.”

“For me.”

“For you.”

Whitfield sat down on the stool very slowly, in the manner of a man who wishes to savor something and is prepared to take his time about it.

“Daniel. It says *see Errol*.”

“It does.”

“There is no possible arrangement of those characters that says anything other than *see Errol*. It is the word *see* and the first letter of the name *Errol*, which are the two words a man writes when he intends to see Errol.”

“One of them says if it were just Errol I’d have written Errol.”

“*If it were just Errol I’d have written Errol*,” Whitfield repeated, with the delight of a collector. He got out a pen and wrote it on the corner of a printout. “I want that. I’m having that.”

“Errol—”

“Do you know what I did on the twentieth of January? I found a thermometer that was eleven hundredths out. Eleven hundredths. It took me a minute and a half and you were sour about it for a week.” He was smiling in a way that had nothing good in it. “That’s the great secret. That’s what four hundred people have spent the winter on. A tired man wrote himself a note to ask the old man about a thermometer.”

“They’ll get bored.”

“They will not get bored,” Whitfield said, and the smile went. “That is the one thing they will never do. Boredom requires there to be an answer, and they have arranged it so there can’t be one.”

The post that Cho read on the nineteenth of February was seven paragraphs long and he read four of them.

It was from the man who ran the board, and it was about doing rather than reading. It said, in the careful institutional prose these people used when they were being serious, that the group had spent eleven years asking other men to show them their data and had never once produced any of its own, and that this was the year that changed. It said there was a member with a working cell — real palladium, real heavy water, real calorimetry, a rectifier out of a plating shop — who had got the loading to hold in October, which was a thing nobody in that group had ever managed.

And it said that the one procedure never attempted in the open literature, the one the record showed had been specifically warned against, would be attempted, and documented, and witnessed.

Cho read that and felt the flat spike of alarm go through him and then subside inside about four seconds, because he thought it through.

The first thought was: that’s a bomb. That’s not an experiment, that’s a pressure vessel full of deuterium and oxygen, and any undergraduate who has been through my lab could tell you so in one sentence.

The second thought, which arrived on top of the first and put it out, was: they can’t.

Because sealing a cell only does anything at all if you can load first, and loading is the part nobody can do. Loading is where thirty-five years of this field has gone to die. Fleischmann couldn’t hold it reproducibly with a proper lab around him. Cho himself, with two years, four hundred thousand dollars, a bath holding to a couple of hundredths and a postdoc who was better at the bench than he was, ran his cathodes at loadings that a 1990 paper would have called unremarkable. It was the hardest thing in the whole business and it was hard for reasons of surface chemistry that no amount of wanting had ever once gotten around.

Real palladium, real heavy water, a rectifier out of a plating shop. A man in a garage.

He was not being arrogant. He would think about that later, at some length, in a room with a detective, and he would still not be able to find the arrogance in it. He had simply applied the thing he knew best in the world to the facts in front of him and concluded, correctly, that a man working alone at a bench with a secondhand power supply was not going to achieve high loading in a heavy-water cell.

He was four months out of date. In a garage two states west a man had asked a plain question at eleven o'clock on a September night and been given a plain answer about surface preparation, and had gone out and looked at his wire under the light, and had fixed it in an hour.

Cho closed the tab and went back to the paper he was writing, which was going well.

Chapter 14 — The Witnesses

They came on the Friday, and not one of them was the person Dale had been reading.

ExcessHeatKev — who had run a board for eleven years, who wrote as though he expected to be quoted, who had told Dale in November that in eleven years nobody had brought them a primary document — turned out to be a slight, soft-spoken man of about fifty in a windbreaker, who came up the driveway holding a rolling suitcase by the handle instead of pulling it, and who said, “Is this all right? Is this an all right time?” three separate times before he got to the garage door.

“You’re Kev.”

“I’m Kev.” He shook Dale’s hand with both of his. “I want to say — I’ve never done this part. The coming. I’ve never come anywhere.”

rwhitcomb was seventy-two and enormous and had driven up from the Gulf in a diesel truck with a toolbox bolted in the bed, and he did not shake anybody’s hand or say hello. He walked into the garage, stood in front of the bench for about forty seconds, and said:

“Who put that fitting on.”

“I did.”

“That’s a compression fitting on a barb.”

“It’s held since August.”

“It’s *held*,” rwhitcomb agreed, with enormous contempt, already opening the toolbox. “Sure. It’s held. My daddy’s Buick held.”

Dale had never been happier to be insulted in his life.

The Tollivers were Jerry and Dean, forty and forty-two, and they had driven eleven hours from Michigan and had brought, in a fitted foam case, a machined stainless flange with an O-ring groove cut to two thousandths, which was the single most beautiful object anyone had ever brought into that garage and which did not fit anything.

“It’s for the cap,” Dean said.

“My cap’s PTFE.”

“Right.” Jerry looked at it. “So we made this for a — we didn’t know what the cap was.”

“You didn’t ask him what the cap was.”

“*You* didn’t ask him what the cap was.”

They put it on the shelf. It sat there through everything that followed, and when it was over it was one of the few things anybody found in one piece.

And Deuteronomy was a woman.

She was sixty-three and about five feet tall and her name was Ruth Amsler, and she got out of a Corolla with a nine-by-thirteen pan in her hands, and Dale, who had spent years reading four-screen posts from her that opened *Friends*. *FRIENDS*. and closed with a benediction, stood in his own driveway with his mouth slightly open.

“I made a hot dish,” she said. “I didn’t know if there’d be food.”

“Ma’am—”

“Ruth. It’s Ruth.” She would not look at him for more than a second at a time. “I write a great deal more than I talk. I’m aware it’s a discrepancy.”

She spoke perhaps forty words that entire first evening. Twice, when Kev was explaining something to the Tollivers, Dale saw her open her mouth and then not, and both times he could see the paragraph go by behind her eyes and get folded up and put away.

Nadine came home at six-fifteen to five vehicles on the street and a garage with the light on and voices in it.

Dale met her at the back door. He had thought about this all week and had not landed anywhere.

“There’s people here.”

“I can hear people, Dale.”

“They’re from the — I know some people who do the same kind of thing. From online. They’ve come to see the setup.”

Nadine looked past him at the garage for a moment. Then she looked at the Corolla, and at the diesel truck with a Louisiana plate, and back at him.

“How long have they driven.”

“Whitcomb’s come from the Gulf. The brothers did eleven hours.”

“Eleven hours,” she said.

“Yes.”

She put her bag down on the step and stood there with her coat still on, and Dale watched her arrive at something and could not tell what it was until she said it.

“Has anybody eaten?”

“Nadine—”

“Because it’s twenty past six and if a man drives eleven hours he ought to get something that isn’t out of a bag.” She was already going in. “There’s a woman out there, I saw her. Bring her in here, she doesn’t want to stand in a garage all night with five men.”

She fed all six of them at a table that seats four, with two folding chairs from the basement, and Ruth Amsler helped her carry things and the two of them worked out inside four minutes that they had both spent twenty years scheduling other people’s appointments, and that was the whole rest of the evening — the two women at one end talking about no-show rates and the men at the other end talking about O-ring compression, and Dale sitting between them thinking that he had not had six people in this house since his mother died.

At half past nine Nadine put a hand on the back of his chair on her way past.

“They’re nice,” she said, quietly, near his ear. “Every one of them is nice. That’s not the part I’d worry about.”

The argument happened on the Saturday morning, and it happened with Sethi on a laptop propped against the flange nobody could use.

He would not fly. He had said so in November and he said so again now, from a kitchen in Alberta at seven in the morning his time, with a mug he kept picking up and not drinking from.

Sethi: I want to say the whole thing once and then I'll shut up and you can all do what you like.

Sethi: Dale, you have built the best amateur cell I have ever seen. I mean that. Your loading is better than mine ever was and I had a machine shop. So hear me as somebody who thinks you're good at this.

Sethi: If you seal that cell you are building a bomb. Not a metaphor. A pressure vessel containing a stoichiometric mixture of deuterium and oxygen with an ignition source inside it. That is the definition of an explosive device. The only reason it hasn't gone off in six months is that you've been venting it.

"I know that," Dale said.

Sethi: Then say it back to me.

"It's a two to one mix of D₂ and O₂ in the headspace and it'll go off across just about any composition, and the electrode's a spark and the palladium's a catalyst so I've got two ignition sources whether I want them or not." Dale said it flat, in order, the way you say back a procedure. "It's not a maybe. It's a when."

There was a pause on the laptop.

Sethi: ...you actually do know.

"Then ask me the other one," Dale said.

Sethi: what are you going to measure

“The bath.” He put two fingers on the jacket. “Thermocouple in the cap, one in the jacket, and I know what that bath does on a cold night because I’ve sat with it since August. I seal it, the loading goes where it can’t go while it’s venting, and if there’s anything in this at all it comes off as heat, and the heat comes off in there.”

Sethi: how much heat

“Two hundred watts is a hundred-watt bulb and another one just like it.” He said it like a part number. “If it makes that on this bench I’ll have it inside ten minutes, and so will everybody standing in this garage.”

Sethi: and if it makes nothing

“Then I’ve got a jar of warm water and I’ve wasted your Saturday.”

Sethi: Then what in God’s name are you doing.

And Dale said the thing that he had been building for four months and had never once said out loud, and the garage went quiet for it.

“Because that’s the whole point of it, Sethi.”

He put his hand flat on the bench beside the cell.

“Everybody in this room has spent years being told there’s nothing there. Thirty-five years of *nothing there*. And then the one time anybody gets a straight answer out of anything, the answer is *don’t do this specific thing*.” He looked around at them. “Not *that won’t work*. Not *don’t waste your time*. It has told me a dozen times that it won’t work and I’ve read every word of it and it’s never once flinched. But you ask it about sealing and it says *please*.”

Nobody said anything.

“You don’t warn a man off a thing that does nothing. There’s no danger in nothing.” Dale’s voice had not gone up at all. “The hazard is the confirmation. The hazard is the only honest thing anybody’s ever said to us, and every one of you knows it, and that’s why you drove eleven hours.”

Kev, quietly: “That’s right.”

Sethi: No. NO. Dale, listen to what you just said. You just told me that the reason to do it is that it’s dangerous. That is the reasoning of a man who has decided already.

Sethi: It is dangerous because it’s a fuel-air mixture in a jar. That is ALSO why they warn you off it. Both things are the same fact. You have taken one fact and made it into evidence for a second thing that isn’t there.

“Then explain to me why the warning is the one thing they wouldn’t say twice.”

Sethi: It said it twice because it says it to everyone. That’s what a safety line IS.

“It said it to two people in two states, word for word, and one of them was a postdoc making a joke and one of them was me.”

Sethi: YES. Because you both asked the same question.

Dale stood there with his hand on the bench.

And the thing that happened next was not an argument, because there was no argument left to have; it was five people in a garage in late February slowly ceasing to look at a laptop. Jerry Tolliver went and got coffee. rwhitcomb picked up a wrench and put it down. Ruth Amsler folded her hands and looked at the floor and said nothing, and Kev said, gently, to the screen, “We’ve heard you, Sethi. We have. It’s Dale’s bench.”

Sethi: It’s Dale’s bench.

Sethi: Dale. Do not stand near it.

Sethi: I'm serious. Whatever else. Long leads, out the door, behind the wall of the house, and you do not go back in for at least an hour no matter what it does or doesn't do. Promise me that and I'll get off.

"I'll do all of that," Dale said. "I'd have done all of that anyway."

He meant it. That was the thing. He meant every word of it, and he was not being reckless, and he had already bought the leads.

Sethi: ok

Sethi: ok.

He signed off and did not come back on.

They spent the rest of Saturday doing it properly, which was Dale's word and which he used four times.

rwhitcomb redid every fitting on the rig with a critical running commentary. They ran forty feet of lead out the side door and around the corner of the house so the supply and the readout could sit on the picnic table in the back yard, behind the brick, out of line of sight of the garage window. Jerry Tolliver, who had done sheet metal, built a three-sided shield out of quarter-inch plate off the truck and bolted it to the bench.

They put the cell inside the shield. They cut the volume down as far as Dale could get it, because a smaller headspace is a smaller charge and everybody in that garage knew it. They checked the cap. They checked the seal three times.

And Kev set up a tripod.

"We document it," he said. "That's the whole thing. That's the part we've never done."

The tripod had come from a store and still had a sticker on the leg, and Kev spent an hour and a half getting it level and could not get the phone to see anything but the shield, which was the point of the shield, and at midnight

Dean Tolliver said, “Kev, the best case here is a video of a piece of steel,” and Kev said, “Then that’s what we’ll have,” and kept adjusting it.

Ruth Amsler wrote down the times. She had brought a notebook for it. She wrote down when they started and when they stopped and who did what, in a small careful hand, and she showed Dale the page at the end of the night and said, “Is that right? Is that how you’d want it?”

“That’s exactly how I’d want it.”

“I thought there ought to be a record,” she said. “Since the whole trouble has been records.”

They sealed it on Sunday morning at twenty past nine.

Dale did it himself. Nobody suggested otherwise and nobody offered, and he would not have let them. He shut the vent line with a quarter turn — a quarter turn, that was all it was, eleven ounces of pressure on a valve handle he had cleaned with a rag the night before — and then he closed the shield and walked out of the garage and around the corner of the house to the picnic table where the other five were standing behind the brick, and he sat down and put his hands flat on the table and looked at the readout.

It was cold. There was still snow in the shade along the fence line. Nadine had gone to her sister’s at eight because she had said, without any edge at all, that she did not want to be in the house today, and Dale had said all right, and that was the entire conversation.

The current was steady. The pressure came up as it should.

“How long,” Ruth said.

“An hour,” Dale said. “Minimum. That’s what I promised Sethi.”

And they sat in a back yard in February and watched a number, six people in coats around a picnic table with a plate of Ruth Amsler’s leftovers on it under foil, and for fifty minutes absolutely nothing happened at all.

Chapter 15 — The Hour

For fifty minutes it was the best morning any of them had ever had.

That was the thing none of them could ever explain afterward, to a detective or a wife or a reporter or themselves. Six people in coats around a picnic table in a back yard in late February with the snow still lying in the shade along the fence, and a number on a readout that was doing exactly what it was supposed to do, and nothing whatsoever to do for an hour.

They got warm on it. Ruth Amsler had brought the leftovers out under foil and nobody was hungry and everybody ate. Jerry Tolliver went to the truck for a folding chair for Ruth and came back with two, and rwhitcomb lowered himself into the second one, which complained about it.

“Now this here,” he said, “is the whole job. People think the job is the doing. The job is the sitting.”

“Is this the refinery one,” said Dean.

“It is the refinery one.”

“He’s told me this twice.”

“Then go stand over there.” rwhitcomb got himself arranged. “Eighty-eight. Gulf Coast, I’m not going to say which. We’ve got a weep on a flange on a hot line — hydrocarbon, three hundred and some degrees, and a weep is a weep until it isn’t. So the operators shut the unit and back everybody off a hundred feet and we’re standing around exactly like this, in the sun, eating, waiting on it to cool enough to get a hand on.”

“How long,” Kev said.

“Four hours is what it needed. And about forty minutes into the four hours, out comes a company man named Ordoyne.”

Dean was already laughing.

“In a *tie*,” rwhitcomb said. “Hundred and one degrees and a necktie and a clipboard, and Ordoyne has been sent out from the office to find out why a unit’s down on a Thursday, and he gets the story from the lead operator and he says — and I heard him say it — he says, *that’s not leaking*.”

“Was it leaking?” said Ruth.

“Ma’am, that is the question.” He held up one finger. “So the operator says it’s leaking. And Ordoyne says it isn’t. And there is not one thing in this world that will move a man in a tie off a position he has taken in front of eight men in coveralls. So Ordoyne tucks that clipboard under his arm and he walks the hundred feet, and he walks up to that flange, and he puts his hand out flat about a foot off it, and he holds it there, and he turns around and he looks at all of us.”

“Oh no,” said Jerry.

“And then he walks back. Slow. Takes his time about it. Gets back to us and says, *it’s not leaking*, and he writes something on the clipboard and he gets in his car and he drives to the office.”

There was a pause.

“And?” said Kev.

“And it wasn’t leaking,” rwhitcomb said. “He was right. Flange was fine. It was a bad gasket sweating and it never was going to let go, and the operator was being careful and the operator was wrong, and Ordoyne was right, and they had that unit back up by six.”

“So he was right.”

“He was *right*.” rwhitcomb leaned forward and put a hand the size of a dinner plate flat on the picnic table. “He was right, and he had no earthly way of knowing he was right, and if he had been wrong there would have been nothing left of him to put in a tie. That’s the story. That is the entire story. I’ve told it for thirty-six years and every time somebody says *so he was right*, and every time I say yes, and it don’t matter even a little bit.”

He sat back.

“We had a hard hat with his name on it in the shop till the day I retired,” he said. “Empty. Hung on a nail.”

Ruth Amsler laughed out loud, once, and put her hand over her mouth, and everybody looked at her, and she went red, and Dale thought it was the best thing he had heard all winter.

Kev checked the tripod through the side window twice and then stopped, because there was nothing to check; it was a phone on a stick pointed at a steel plate, recording a steel plate, and it would record a steel plate for as long as the battery held.

“It’s the principle,” he said, when Dean said something about it.

“I know it’s the principle.”

“Eleven years of asking other men for their data.” Kev had his hands around a cup he wasn’t drinking. “I’d rather have four hours of a steel plate that’s ours.”

Ruth wrote the times down. **9:20 sealed. 9:26 current steady. 9:41 P rising, steady rate. 9:55 no change to report.** She showed Dale the page and he said it was exactly right, and she looked pleased, which she did not often have cause to be about anything.

At ten o’clock Dale said, “Half past ten’s the hour,” and nobody answered because nobody had been counting and everybody had.

At six minutes past ten the readout stopped making sense.

Not dramatically. All morning it had gone up a straight boring line, and at ten o’clock it had read a shade over fourteen. Now it jumped — 14.1, then 9.8, then 15.6, then 3.1, then back up. The current reading did the same thing a few seconds later.

Dale looked at it for about forty seconds.

“What,” said rwhitcomb.

“Lead’s noisy.”

“Or the gauge is done.”

“Or the gauge is done, or there’s water in the connector, or I’ve got the shield sitting on the cable.” Dale said it in the flat voice, in order, as he had said the hazard back to Sethi. “It’s forty foot of twenty-two gauge I bought in a hardware store and ran out a door onto wet grass. It’d be a wonder if it was clean.”

“So it’s the wire.”

“It’s almost certainly the wire.” He kept looking at the numbers. “That’s the trouble with it.”

Because if it was the wire, then he had no reading at all. And if he had no reading at all he had a sealed vessel forty feet away with gas accumulating in it and no way to know how much, which was the one condition in the entire world that he had built the whole morning to avoid.

He stood up.

“Dale,” Kev said.

“I’m going to look at the gauge.”

“He said an hour.”

“He said an hour if it does something or doesn’t do something. He didn’t say an hour blind.” Dale was already moving around the end of the table. “I’m not going to stand out here for twenty-four minutes with no numbers on a thing I sealed. That’s not caution, Kev, that’s just not looking.”

“Then let one of us—”

“No.” He said it without any force at all, which was worse. “It’s my bench.”

Ruth Amsler said, “Dale.”

He stopped.

She had her pen in her hand and the notebook open on her knee and she did not look up at him, and it took her a moment, and it was the longest sentence she said out loud the entire time any of them knew her.

“I would rather have no record at all than have written down what time you went in there.”

Dale stood in his own yard and looked at a small woman in a folding chair with a notebook on her knee.

“It’s twenty feet, Ruth,” he said gently. “I’m going to read a dial and come back out. I’ll be forty seconds.”

He went around the corner of the house.

The garage smelled as it always did, hot dust off the transformer and a little solder and the cold coming in under the side door.

He did not hurry, because you do not hurry, and because after thirty years his body knew the difference between working quickly and hurrying the way it knew the difference between a good weld and a pretty one.

He stopped six feet inside the door and looked at the whole room first. That was training. You look at the room before you look at the part.

Everything was ordinary. The bench, the light, the shelf with the Tollivers' beautiful useless flange on it. The cell inside the three-sided shield with the plate between it and the door, which meant he was going to have to come around the open side to read the gauge, which he had known walking in and which was fine because he would be there and gone.

He saw one thing he did not like and it was not the cell. The lead ran out the side door and he had laid a folded towel over the threshold so the door would sit on it without pinching, and the towel had gone dark. Wet. Snowmelt off the roofline had run down the door and into it, and the cable was lying in it.

There it is, Dale thought. That's your gauge.

He felt the whole morning settle back into place. It was the wire. It had been the wire the entire time, it was a wet connector on a hardware-store cable, and he had a spare pigtail in the drawer and it was a four-minute job and then he would have his numbers back and he would go out and sit in the cold with Ruth's notebook and wait out the rest of the hour like he'd said he would.

He was, at that moment, entirely happy.

He came around the open side of the shield to read the dial with his own eyes, because a man reads the dial with his own eyes, and he put his left hand on the corner of the bench the way he had put it on ten thousand benches.

The gauge said forty-one.

He had about a fifth of a second to understand that number, and he did understand it.

He had been right about the wire. He had been right about the towel and the snowmelt and the connector, as right as he had been about Tuck's gasket, as right as he had been about anything in thirty years — and a bad wire had never once meant the jar was all right. It had only ever meant he could not see the jar. He had known that outside at the table, and he had come in here to fix the part he could fix.

That was the last thing Dale Furlong thought.

Behind the house, what they heard was not a bang.

Four of the five described it later, separately, to a detective, in almost the same words: a flat hard *crack* with no boom under it, over before it started, and then the sound of the garage — the roof coming up an inch and setting back down, the door bowing outward and every window in it going at once, glass going out across the driveway in a sheet.

Then a second of nothing. Then the noise of things falling that had not finished falling.

Jerry Tolliver was up and moving before the glass landed and rwhitcomb had him by the coat before he got four steps, one enormous hand in the collar, and held him.

"You wait."

"He's in there—"

"You *wait*." rwhitcomb had spent thirty years on units that go up. His voice had nothing in it at all. "There's a second one in near every one of these and it's the one that gets whoever ran in. You count sixty."

They counted sixty. It was the longest thing any of them ever did.

Kev had his phone out and could not make it work, and then made it work, and gave the address wrong twice and then right, and stayed on the line, and the dispatcher's voice was very calm and asked him things he could not answer, like whether there were chemicals in the structure, and Kev

said, “Heavy water,” and the dispatcher said, “Sir, is that a chemical,” and Kev said, “No. No, it’s — no.”

Dean Tolliver got the hose bib on and stood there holding a garden hose in February with no idea what he intended to do with it.

And Ruth Amsler sat down on the folding chair, because her legs had stopped, and she opened the notebook on her knee and looked at it for a while.

Then she wrote **10:14** and put the pen down on the table and did not write anything else in it, that day or ever.

Nadine was at her sister’s with a cup of coffee she had been holding for twenty minutes when the phone went.

She had not been able to settle all morning. Frances had asked her twice what was wrong and she had said nothing was wrong, and the second time Frances had let it go.

She looked at the screen. It was not Dale.

It was a number from the county with a 911 dispatch tag on it, and Nadine Furlong sat in her sister’s kitchen with the coffee going cold in her hand and looked at it ringing on the table and understood — before she picked it up, in the space of about a second and a half, with total and irreversible clarity — every single thing that had happened in her own house since a Tuesday in July when a charge for eleven hundred and forty dollars came through her phone while she was on hold with a pharmacy.

She picked it up.

“This is Nadine,” she said.

Chapter 16 — Cooperative

Terri Ackerman had worked nineteen fatalities in twenty-two years and had never once had a scene where everybody stayed.

That was the first thing she wrote down, standing at the end of the driveway at one in the afternoon with the fire marshal's people still in the structure. Five witnesses. Not one of them had left. Two were in an ambulance being looked at for glass and would not get in it properly, sitting sideways with their feet out the back so they could see the house. One was sitting on the tailgate of a Louisiana truck with a blanket he hadn't asked for. One was standing at the tape with a phone held out toward her, waiting, the way you wait to hand someone a wrench.

The fifth was a small woman in a folding chair somebody had carried around from the back yard, holding a spiral notebook on her knee with both hands.

Nobody had gotten a lawyer. Nobody had gone home. By two o'clock she had five statements, all of them volunteered, all of them consistent, and three of the five had apologized to her for wasting her time.

Cooperative, she wrote, and then looked at the word, because in twenty-two years she had learned that when everybody is helpful it means one of two things and only one of them is good.

The garage had come apart from the inside.

That much the marshal's man had before she got there and it wasn't hard: the roof had lifted off the plate and reseated, the door had bowed outward, the glass was on the driveway and not in the room. A three-sided steel shield bolted to the bench had done a good job of protecting the wall behind it and nothing at all for the open side.

"Somebody built that," Ackerman said.

"Somebody built all of it." The marshal's man, Delgado, was crouched by the bench with a light. "That's the part that's going to be strange in the report. This is not a meth thing and it's not a still and it's not a guy huffing something. This is bolted, grounded, labeled. There's a shield. There's forty

feet of lead run out the door so he could watch the readout from behind a wall." He looked up. "He did everything right."

"And then?"

"And then he was standing next to it." Delgado stood, knees popping. "Best I've got right now is a pressure vessel with a flammable mixture in it. There's a valve on the vent line and the valve is shut. Whether he shut it or it failed shut, I'll tell you in three weeks."

"He shut it," Ackerman said. "There's five people out there who watched him do it at nine-twenty."

Delgado looked at her.

"Well," he said. "That'll shorten things."

She interviewed all five that afternoon and into the evening and it was the strangest four hours of her career.

Kevin Slate, 51, Ohio. Slate had a printed timeline. He handed it to her within about ninety seconds of introducing himself and it was accurate, and it matched, in every material particular, the four accounts she took afterward and had not yet heard. He also had a phone on a tripod that had recorded four hours and twenty minutes of a steel plate, and he wanted her to have it, and he asked her twice whether she was sure she had the whole file.

Ellis Whitcomb, 72, Louisiana. Retired instrument technician, thirty-six years at three refineries. Gave her the clearest account of the five, in order, with times, and used the word *deflagration* and then apologized for using it. He was the one who had physically restrained another witness from running toward the structure and had made them all wait sixty seconds, and when she said that had probably been the right call he said, "It's not a call. It's a rule," and looked away.

Jerry and Dean Tolliver, 42 and 40, Michigan. Brothers. Machine shop. Interviewed separately and told the same story with the same two details out of order.

Ruth Amsler, 63. Two hours from here.

Amsler gave her the notebook.

Ackerman had not asked for it. The woman took it off her knee and held it out with both hands and said, "I was keeping the times. I don't know if that helps."

It was a school notebook. In a small controlled hand it said:

9:20 sealed. 9:26 current steady. 9:41 P rising, steady rate. 9:55 no change to report.

And then, in a different pressure of pen, three lines lower:

10:14

Nothing after it.

Ackerman looked at that page for longer than she meant to.

"Ma'am, I'm going to need to keep this."

"I know," Ruth Amsler said. "That's why I brought it out."

What every single one of them wanted to talk about, and what none of them could be steered away from for more than four minutes at a stretch, was a laboratory two states east.

She got the whole architecture of it by six o'clock and she got it from five people who were trying very hard to be clear. There was a physicist. There had been a reading. There was a machine that had said something, and a document with a university seal on it, and a schedule with a hundred and six lines in it, and a notebook.

"He drove out there," Slate said. "In October. He took a tour."

"Furlong did."

"He took a public tour. He filled out a form." Slate had his hands flat on the table. "He didn't lie to anybody. I want that in what you write down. Dale never once lied to anybody about anything."

"And the notebook?"

The pause was about a second and a half, and she had been doing this long enough to know exactly what a second and a half is worth.

“He brought it back with him,” Slate said.

“Did anybody at that university know he had it?”

“Dr. Cho knew Dale had been there. They spoke.” Slate said it without any hesitation at all, which she also noted. “In the hallway. That same morning, on Dale’s way out.”

“How do you know that?”

“Because Dale told us that night. All of it.” He had his phone out already. “He wrote down what Dr. Cho said to him and what he said back. He thought it was important.”

He turned the phone around and she read it twice.

dfurlong: he came round the corner off the stairs and stopped me.
asked was I one of the people who’d been writing to him

dfurlong: I said I filed a records request. he said my name back to me

dfurlong: he told me flat there’s nothing there and then he told me
why they held the 106 and I think he meant every word of it

dfurlong: I told him I believe you. and then I told him I believe that’s
what you know

“Mr. Slate,” Ackerman said. “When he typed that, where was the notebook?”

Slate looked at the table.

“In his truck,” he said.

They pulled it out of the wreckage on the Tuesday, and it was in better shape than anything else on that bench, because it had been under a shelf and the shelf had come down flat on top of it.

She already knew what it was. She had known since about half past four on the Sunday, because five people had told her, in order, without being

pressed: a bound laboratory notebook belonging to a Dr. Daniel Cho, carried out of a physics building two states east in October by a man who had booked the tour on a form and signed his own name to it. She knew whose hand the entries were in and whose hand the marks on top of them were in, and she had not had to work out either.

So she was not in the evidence room on Tuesday to find out what it was.

She was there for two hours to find out whether there was anything in it — because that was the only question that decided what kind of case this was. If the book was nothing, then a decent man had killed himself over a misunderstanding and she would write it up as one. If the book was something, then four months ago a university had lost a document it never reported losing, and the reasons for that would need to be somebody's problem.

It was a composition book, and on the inside cover in a small square hand was a legend:

GREEN — entry contains a number. BLUE — entry contains no number. RED — gap in dates. ORANGE — hand changes.

Every page after it was marked. Four colors, ruled straight, corner to corner, four hundred entries across two years. Somebody had gone through the entire thing with a straightedge and a system.

She was not a scientist and she did not pretend to be. But she had spent twenty-two years reading what people do when they are working something out, and she knew a man's own handwriting from a man's annotations, and there were two hands here and only one of them was making the marks.

The underlying entries were dense and abbreviated and she could not read them. *BL run 4. dP -0.02. eq. RC ck.* A private shorthand.

The marks on top of them were something else. They were *organized*. They were the work of somebody who had found a pattern and was proving it out, methodically, over weeks, in colored ink, in a garage.

Ackerman closed the book and sat back and thought: *a man does not do this to nothing.*

She would come back to that sentence in about two months, on the telephone, to the man it had been aimed at. But on the Tuesday, in the evidence room, it was simply what the object in front of her looked like, and it looked like it for the same reason it had looked that way to Dale Furlong in November — because the marks were careful, and because careful is the thing human beings cannot help reading as meaningful.

The messages came off the recovered phone on Thursday and she read all of them and then read one of them four times.

It was in a private group with eight members and it had a timestamp of 7:41 on the Saturday morning, twenty-six hours before.

Sethi: If you seal that cell you are building a bomb. Not a metaphor. A pressure vessel containing a stoichiometric mixture of deuterium and oxygen with an ignition source inside it. That is the definition of an explosive device. The only reason it hasn't gone off in six months is that you've been venting it.

And Furlong's answer, in the transcript, spoken aloud in the room and typed up by somebody afterward — she had that too, because Slate had typed it up afterward and kept it, because Slate kept everything:

It's a two to one mix of D2 and O2 in the headspace and it'll go off across just about any composition, and the electrode's a spark and the palladium's a catalyst so I've got two ignition sources whether I want them or not. It's not a maybe. It's a when.

Ackerman sat in her office with a cold coffee and read that four times.

He knew. He had been told, in writing, by a competent person, twenty-six hours in advance, and he had said it back correctly, in technical language, and then he had done it anyway.

That was not an accident. Accidents are what happen to people who don't know.

She wrote in her notes: *not negligence. belief*. Then she looked at that for a while and crossed out *belief* and wrote *told him it was worth it?*

On the Friday she called the university.

She got the physics department front office, and then a records office, and then, after some negotiating, a woman named Kovach who was extremely helpful and slightly too fast, and who confirmed that yes, a public records request had been filed in July by a D. Furlong, and that yes, forty-one pages had been released and a hundred and six items withheld under a research-data exemption, and would the detective like the file, because she could send the file today.

“Ms. Kovach, I’ve got a laboratory notebook here with your university’s name inside the front cover.”

The pause on the line was four or five seconds.

“I don’t know anything about that,” Kovach said.

“It’s a bound lab notebook. Two years of entries. Dr. Daniel Cho’s name in it.”

“I understand. I’m telling you I have no record of it.” Papers moved. “Detective, can I ask — was it reported?”

“That’s what I’m calling to find out.”

Kovach put her on hold for fourteen minutes.

When she came back her voice had changed in a way Ackerman had heard maybe two hundred times in her career. It is the sound somebody makes on working out that they are the one who has to say it.

“There’s no report,” she said. “Not with campus police, not with the city, not with our office. There’s nothing. I checked October through last week.”

“So a man walked out of a university building with a professor’s laboratory notebook,” Ackerman said, “and in four months nobody wrote it down anywhere.”

“That’s —” Kovach stopped. “That appears to be correct.”

Ackerman thanked her, and hung up, and sat looking out the window at the parking lot for a while.

Then she got out a fresh page and worked it through in order, as she always did, because the innocent version deserved to be written down first and she had learned a long time ago to write it down properly rather than glance at it.

1. Man loses a notebook. Assumes he mislaid it. Data's backed up anyway. Never reports it because there's nothing to report.

That covered everything, and it covered it well, and it held right up until the second line.

2. Except he met him. In a hallway, that morning. Furlong told him his name and told him he was the one who'd filed the request.

3. So the book goes missing out of that room and there is exactly one man on earth he can name for it.

4. Four months. Campus police, city, records office. Nothing anywhere.

She sat back.

It wasn't the missing notebook. Notebooks go missing; she had been a police officer for twenty-two years and knew perfectly well that most people who lose a thing assume they lost it. It was that this one man had been handed the answer in his own corridor, by the man himself, with his name attached — and had then spent four months not writing a single word of it down.

Why does a man who already knows who took it decline to say so?

She underlined that, which was a habit, and then looked at the page and thought about the four colors of ink in the evidence room, and about a woman holding a notebook out with both hands, and about five people who had not gone home.

She booked the flight for the eleventh of March.

Chapter 17 — The Eleventh of March

She did not call ahead. Cho understood later that this had been deliberate and at the time thought nothing of it at all, because a woman with a county ID in her hand at the door of a physics building on a Tuesday is not a category the mind has anything ready for.

“Dr. Cho? Terri Ackerman. I’m a detective, I’ve come in from out of state, and I’d like about an hour if you can give it to me.”

“Is this about the logbook?”

She looked at him for a moment.

“Partly,” she said.

She told him standing up, in the corridor outside 214, because she had wanted to be somewhere she could see his hands and his face at the same time and because rooms make people careful.

“On the twenty-third of February a man named Dale Furlong was killed in his garage. It was an explosion. He was the only person hurt.”

Cho did not say anything at all.

“There were five other people present,” Ackerman said. “Nobody else was in the structure.”

He put one hand out and rested it on the wall.

That was the whole of it, and Ackerman watched it happen and did not know what she had watched, because there is no reliable difference at four feet between a man absorbing news and a man performing absorption, and she had learned twenty years ago to stop believing she could tell.

“Do you need to sit down?”

“No.” He took his hand off the wall. “Do you — I’m sorry. Was he married?”

“He was.”

“Is she all right?”

“No,” Ackerman said. “But she’s got people.”

Cho nodded slowly, twice, and then he said, “How did it happen,” and Ackerman said, “That’s part of what I’m here about,” and did not answer. So he answered it himself.

“He sealed it.”

Ackerman said nothing.

“He built a cell.” Cho was looking at the floor, and the words were coming out at a speed she had not expected. “Heavy water, palladium wire, some kind of supply. And he sealed the headspace, or he closed the vent line and left it running, and it filled up with deuterium and oxygen, and it went. It’d take —” he stopped, and she could see him doing something behind his eyes, and it took him about three seconds — “not long. An hour. Under an hour if he had any current on it at all.”

“Dr. Cho—”

“Was he standing at the bench?”

“Why do you ask that?”

“Because if he was forty feet away with a wall in between he’d be alive,” Cho said, “and you told me he was the only one hurt, so there were five people somewhere else that were fine, so he was at the bench and they weren’t.”

The corridor was quiet.

Ackerman had come here with a page of questions in her coat pocket, and she had just been given, unprompted, in under forty seconds, by a man who had been informed of a death ninety seconds earlier, the mechanism, the timeline, the reason, and the position of the body.

“That’s a lot,” she said, “very fast.”

And Cho heard it. She watched him hear it — watched the whole thing arrive on him at once — and he closed his eyes briefly and said, “Yes. I know it is.”

“Then help me with it.”

“It’s not fast.” He opened his eyes. “Detective, that’s the *only* thing it could be. There isn’t a second answer. If a man with a heavy-water cell dies in an explosion, he sealed it. That’s not deduction, it’s — it’s the one and only way that apparatus kills anybody, and every person who has ever run one knows it, and it’s on a card by that door.” He gestured, and she looked, and there was a laminated card on the wall by the hood, yellowed at one corner, that said in large type **NEVER SEAL A CELL THAT IS MAKING GAS.** “I teach it before I teach pipetting. I have taught it to about four hundred people.”

Ackerman went and read the card.

“How old is this?”

“Nine years.”

“Somebody’s been at the corner of it.”

“It’s gone yellow,” Cho said. “A man told me to get it replaced. Standing about where you are.”

She got him into a room after that, and he did not ask for anyone and did not ask whether he should, and she noted both, and did not mention that she had no authority in this state to keep him in it.

She took him through it in order and he answered everything. The Friday in June. The postdoc’s joke. The screenshot. The program officer. The clarification. The records request, the exemption, the hundred and six withheld records. He explained the research-data exemption clearly and without complaint and volunteered that he had been offered the chance to waive it and had declined.

“Why?”

“Because our records officer advised me not to and her reasoning was good.”

“What was her reasoning?”

“That if I waived it once, the next requester would cite that I’d waived it.”

“And was there a next requester?”

“There was a federal one in November,” Cho said. “I don’t know who.”

Ackerman wrote that down and moved on, because it was going to check out and she could feel it going to check out, and because in twenty-two years she had learned that the thing you are actually looking for never comes up in the part where they are being helpful.

It came up at the fifty-minute mark.

“Tell me about the twenty-second of October.”

“He came on a tour.”

“You spoke to him.”

“In the corridor. On his way out.” Cho did not hesitate and did not embroider. “I asked him if he was one of the people who’d been writing to me. He said he’d filed a records request. I said his name and he said that’s right.”

“And then?”

“And then I told him there was nothing here.” Cho’s mouth did something. “And he said he believed me. And then he said he believed that was what I *knew*, and he went out the fire door.”

Ackerman had that exact sentence in a dead man’s own writing in a file in her bag, and hearing it come out of this man’s mouth in the same words did something to the back of her neck that she chose not to examine.

“When did you find out the notebook was gone?”

“The Friday. Two days later. My postdoc noticed.”

“And you knew who had taken it.”

“Immediately.”

“You had his name.”

“I had his name.”

“You knew he’d been in the building, you knew he’d been outside that room, and you knew he was the only stranger who’d been on that floor that morning.”

“Yes.”

Ackerman put her pen down, which she did about twice a year.

“Dr. Cho, in four months you did not file a police report, a campus incident report, an insurance claim, a departmental loss form, or an email to anybody outside this hallway.” She let that sit. “Why.”

He told her the truth, and it took him four minutes, and she watched it get worse with every sentence he said.

He told her about the field marked *Description of item(s) taken*. He told her that a report in this state is a public record and disclosable, and that he had learned it the hard way over the summer. He told her the report would have had the words *cold fusion* in it and would have existed forever and would have been quoted within a week.

He told her his year-three release had executed on the ninth of November.

“So you found out on the twenty-fourth of October,” Ackerman said, “and your money came in on the ninth of November.”

“Yes.”

“And you didn’t file before the ninth.”

“No.”

“Did you consider filing after the ninth?”

“Yes,” Cho said. “That’s the part I’d like to say properly, because it’s the worst of it and I’d rather say it than have you find it.” He had his hands flat on the table. “By the ninth I understood that a report filed on the tenth would look like a report filed by a man who’d waited for his money. And I understood that every day after that made it look worse in exactly the same direction. And I understood that at the time. Not that I forgot. I stood in the lab on a Wednesday morning and worked out that the window had shut, and then I went and ordered two pressure transducers.”

Ackerman looked at him for a while.

“Who told you not to write things down?” she said.

“Nobody told me.”

“Somebody gave you that idea.”

And Cho, who had answered every question for fifty-four minutes without once looking away, hesitated for the first time.

“The system,” he said. “Sage. The AI. In September I asked it what I could do about all of this and it told me the only real lever I had was to stop producing documents, because every new document becomes something new for them to read.” He said it flatly, and heard how it sounded, and said it anyway. “It was right. It was right about everything else too.”

Ackerman wrote ten words in her notebook.

asked the machine what to do. it said stay quiet.

She talked to the others afterward and it did not go the way she expected in either direction.

Vega was hostile for about ninety seconds and then not. She cried once, briefly, in the middle of a sentence about a screenshot, and got angry at herself for it, and then answered everything with a precision Ackerman found genuinely useful, including the thing nobody else had thought to tell her, which was that the orange marks in the marked-up notebook — the *hand changes*, the nine of them, the pattern the whole board had turned on — were her own handwriting on days when Cho was out of the building.

“They think that’s the tell,” Vega said. “That’s the big one. Nine times. It’s *me*. I was writing the entry because he wasn’t here.”

Ackerman had not known that, and it sat down in her chest and stayed there.

Whitfield was a catastrophe for ten minutes and then the single most useful witness of her career.

The catastrophe was that he could not answer a question without correcting its terms, and that when she said the word *reaction* he said “there wasn’t one, and that’s rather the entire point,” and that when she asked whether the notebook had contained anything valuable he said, “That depends

enormously on what you mean by valuable, and I'd want to establish that before I said anything you might write down."

Then he asked to see the pictures.

She showed him the marked pages and he went through them for six or seven minutes with his glasses pushed down, saying nothing.

"Ask me what *RC ck* is," he said.

"What's *RC ck*?"

"Reference cell check. Nine days in August. We borrowed an instrument from a vendor and gave it back, and there is an invoice, and I can produce it in about four minutes." He turned a page. "*eq* is equilibrium. It appears after a bath swap, because that is when one waits. *v* is a tick made by a pen that was running out; you can see it fail again at the top of the following page. And *See E.* is *see Errol*."

"Who's *Errol*?"

Whitfield took his glasses off.

"I am," he said. "On the nineteenth of January last year Daniel could not make his standards agree and he ran out of ideas by four o'clock and wrote himself a note to ask the old man in the morning, and in the morning I found a thermometer that was eleven hundredths of a degree out, and I was insufferable about it for a fortnight." He put the glasses in his cardigan pocket. "There are four hundred people who have spent a winter on those two words. One of them worked out that it was me and was told it was too easy."

Ackerman looked at the page.

"It is too easy," she said.

"It is *precisely* as easy as it looks," said Whitfield. "That's the difficulty."

She left at ten past four and Cho walked her down to the lobby, past the wall of framed group photographs going back to 1968.

At the door she stopped.

“Dr. Cho, I’m not going to tell you it’s fine, because I haven’t finished and I don’t say things I haven’t finished.”

“All right.”

“Here’s where I am.” She was buttoning her coat and not looking at him, which he understood was a courtesy. “I’ve got a dead man who was told in writing, twenty-six hours ahead, by somebody competent, exactly what would happen, and did it anyway. I’ve got five witnesses who won’t stop volunteering things. And I’ve got a professor who never reported a theft he could have named a suspect for on the same afternoon.”

“I understand how that reads.”

“I don’t think you do, quite.” She got the last button. “It reads like the notebook mattered.”

She went out through the fire door — push-bar, everyone uses it — and Cho stood in his own lobby and watched her cross the lot to a rental car, and then went back upstairs and sat down at his desk and did not do anything for about an hour.

Chapter 18 — Next of Kin

There were fourteen casseroles in the house by Tuesday and Nadine could account for twelve of them.

That was what she had to work with. Two dishes had come in with no name on the foil, and a woman who had spent twenty years running the week for a clinic that saw four hundred people sat at her own kitchen table on the Tuesday after her husband died and made a list of whose pan went back to whom, and got stuck on two, and stayed stuck for a while.

Frances took the list off her eventually.

“You don’t have to do this.”

“Somebody’s got to get the dishes back.”

“Nadine.”

“That’s a Pyrex,” she said. “People want a Pyrex back.”

The garage had tape on it for four days and then plywood, and then a man from the county came and put a sticker on the plywood, and after that nobody mentioned the garage for two months.

She did not go out there. It was not a decision she made. The side door faced the kitchen window and she stood at that window forty times a day and looked straight through it.

The five of them stayed the week, because the detective had asked them to be available and because none of them could work out how to be the first one to go.

Kev asked on the Monday whether there was anything he could do, and she said no, and he asked again on the Tuesday, and on the Wednesday he was still in the driveway in the same windbreaker, and she understood that the question was not going to stop and that it was not really a question.

So she gave him the list.

There were thirty-eight people who had to be told. Dale’s sister in Florida. The Ferreiras. Tuck at the Ridgeline. A cousin nobody had seen since 2009.

Ostrowski's widow. Eleven men from the plating shop, of whom she had current numbers for four.

"I'll do it," Kev said.

"You don't know one of them."

"No, ma'am. That's why I can."

He did all thirty-eight from her kitchen over two days, with a legal pad, and he was very good at it. He got the four shop numbers up to eight by calling a man who knew a man. He wrote down what each of them said and whether they wanted calling again. He asked her three separate times whether it was an all right time to start.

She stood at the sink with her back to him and listened to him say her husband's name to strangers in a soft voice, over and over, and thought: this is a careful man. This is a man who keeps a record.

Whitcomb left on the Thursday. The only thing he ever said to her about any of it was on Wednesday night, in the kitchen doorway, with his coat already on.

"Anything in the house not working right?"

"The water heater takes a while."

He was under it two hours. He did not tell her what he'd done and he did not say goodbye; she heard the diesel turn over at ten past six on Thursday and by the time she got to the window it was at the end of the street. The water ran hot in forty seconds after that and did all year.

The Tollivers spent Monday on their hands and knees in the front yard picking glass out of the grass, both of them, in February, and would not come in the house and would not eat anything, and at four in the afternoon Dean asked whether it would be all right if they went, and then apologized for asking, and then apologized for apologizing.

They buried Dale on the first of March. A hundred and some people came, most of them from the shop. The man from the funeral home called her *Mrs. Furlong* forty times. Tuck stood at the back and cried, which surprised her.

She went back to work on the tenth.

Frances said it was too early, once, and then let it go. The clinic had told her to take as long as she needed, which is a thing an office says, and she had watched enough people take as long as they needed to know what it does to a schedule.

The first call she took was a woman moving a Thursday to a Monday.

She handled it. Green, because a woman who calls three weeks ahead to move a Thursday always comes. She finished the note and put the phone down and got up and went out to her car in the lot and sat in it with her hands in her lap for ten minutes, and then went back in and did the rest of the day.

The thing she could not get around, and she got at it from every side for about a month, at the sink and in the car and at four in the morning, was that she had declined to ask.

Eleven years ago a man at a trade show had sold Dale a case of magnets that clamped onto a fuel line. She had said nothing about it then and nothing about it since. That had been the correct call and she had never once doubted it — asking would have cost them a bad month and would not have brought back nineteen hundred dollars. She had made a rule out of it, and she had kept the rule the way you keep a rule, which is on the days it costs you something.

So in July there was a charge for eleven hundred and forty dollars to a metals company in New Jersey, and she asked him what it was for, and he told her the truth as far as he was willing to carry it, and she asked him one more question and got a straight answer.

Is anybody getting hurt by this?

No.

She had read that man correctly for thirty-one years. She read him correctly in the kitchen doorway in July — a man doing something real and telling her a smaller version of it, which was how he answered everybody about everything. What she had not done was ask the second question. She went

over that for a month and at the end of the month she still could not build the version where she asked it and he told her.

Everything she knew about what her husband had spent eight months doing, she had been told by people she had fed once.

Ruth Amsler wrote first.

It came on the eighth of March, four pages, both sides, in a hand that leaned hard to the right, and it was the only thing anybody sent her that she read all the way through.

Dear Mrs. Furlong,

I have begun this four times and each time what came out was more about me than about you, and I have thrown those away, and I would like to try once more.

You fed six people at a table for four. You brought me in out of the garage without being asked. I noticed it at the time and I have thought about it every day since.

I was keeping the times that morning. I have given the notebook to the detective, which I would do again. I will tell you everything that is in it if you ever want it, and I will never mention it again if you don't.

I am two hours away and I do not require anything from you.

— Ruth Amsler

She came at the end of March, on a Saturday, because Nadine finally wrote back four words, and she brought a hot dish, because that was what she had.

They sat in the kitchen an hour and talked about the clinic. No-show rates. A girl Ruth had trained in 1998 who was running a whole office in Duluth now. Ruth was easier at a table than she had been in a garage, though she still would not look at anybody for longer than a second.

“Ruth.”

“Yes.”

“I want the hour.”

Ruth put her cup down.

"All of it," Nadine said. "From when I left the house at eight. I've had it from the detective and I've had it from Mr. Slate and both of them gave it to me like a report."

"All right," Ruth said.

Then she talked for twenty minutes, which was more than she had said out loud in the entire weekend she spent in that house, and she did it in order and with the times in it, because she had the times.

The picnic table. Snow still along the fence in the shade. Jerry going to the truck for a folding chair for her and coming back with two. The foil off the leftovers and nobody hungry and everybody eating. Mr. Whitcomb telling a story about a refinery in 1988 and a company man in a necktie who put his hand out flat a foot off a hot flange, and Ruth laughing out loud in a back yard and putting her hand over her mouth. Kev out at the side window checking a tripod that had nothing to check, because it was a telephone on a stick pointed at a sheet of steel, and Kev knowing that, and going out to look at it anyway.

"He was happy," Ruth said.

"He was happy."

"Mrs. Furlong, I have never seen anybody that happy. He had six people in his yard and every one of them had come a long way and he had built the thing himself and it was doing what it was supposed to do." She turned her cup a quarter turn on the table. "For fifty minutes it was the best morning any of us ever had. That's what I can't put anywhere."

Nadine got up and put the kettle on, which did not need putting on.

"Was he scared?"

"No."

"At the end. Going in there."

"No," Ruth said. "He was pleased. He'd worked out what was wrong with it and he was going to go read the dial and come back out. He said forty

seconds.” She had her hands flat on the table now. “I told him I would rather have no record at all than write down what time he went in there. I said that out loud, to him, and he said it was twenty feet.”

The kettle started.

“And then Mr. Whitcomb wouldn’t let Jerry go in,” Ruth said. “He had him by the coat and he made us all count sixty. I counted out loud. I still don’t know why I counted out loud.”

Before Ruth left, Nadine said, “There’s a case of magnets in my basement.”

Ruth waited.

“Under the stairs. Eleven years. Two dozen of them in blister packs with a picture of a fellow smiling at a gas pump. You clamp them on the fuel line and they’re supposed to get you four more miles to the gallon.” She was rinsing a cup she had already rinsed. “Nineteen hundred dollars. And I have never said that word out loud in this house. Not once in eleven years, and he knew I never had, and that was the whole arrangement.”

She shut the tap.

“I’d do it again,” she said. “I want that understood by somebody. It was the right thing every single day I did it.”

Ruth Amsler stood in the kitchen with her coat over her arm and did not say anything for a while.

“I’ll know it,” she said.

Kev sent a padded envelope in the second week of March with a note on the front saying he thought she ought to see it before strangers did.

The first page was a post he had put up on the second and pinned to the top of the board.

ExcessHeatKey: Dale Furlong did the work. In eleven years nobody on this board has done it. He built the cell, he got the loading to hold, he

kept a record, and he never once misrepresented anything to anybody, including me.

ExcessHeatKev: On the Saturday morning a competent man told this group exactly what was going to happen and told us why. I heard him say it. I answered him in five words. I said it's Dale's bench. Then I went and stood behind a wall.

ExcessHeatKev: I'm closing the private group and I'm stepping down from the board. Not because I've stopped believing anything. Because I was the one who could have stopped it, and I didn't. I don't get to be the one who decides after that.

Behind it, on a separate sheet, was something nobody had asked him for.

ExcessHeatKev — *private*

Sethi. You don't have to read this and you don't have to answer it.

You told us twenty-six hours ahead. You said the whole thing plainly and then you said it again, and I have the log, and I've read it about forty times now.

I said we've heard you. I want you to know what that sentence was. It wasn't disagreement. I understood every word and I couldn't find one thing wrong with any of it, and I said we've heard you anyway, because there were five people in a garage who had driven a long way and I did not want to be the man who stopped it.

That's the whole of it. That's what I've got.

I'm not asking you for anything. I'd rather you heard it from me than read it on the board.

And under that, in ballpoint, in Kev's own hand:

Sethi never answered. I don't believe he's going to. I don't blame him and I'd like you to know I don't.

Nadine read the second page twice and the first page once.

What she noticed, and did not say to anybody, was that the only complete and truthful account anyone had produced of what happened in her own house was addressed to a man in Alberta.

She put the whole thing in the drawer with the insurance paperwork.

Kev wrote every month after that. He never asked her for anything and he never mentioned the board again, and around the fourth one she started writing back.

She asked it herself on a Tuesday in April, in the car in the clinic lot, with twenty minutes and a sandwich she wasn't going to open.

It had been explained to her three times by then. The detective had explained it and had been careful and kind and had used the word *deflagration* and then explained that too. Mr. Slate had explained it in a letter with a diagram in it. Ruth had explained the part she saw. All three of them had told her the truth and she had come away from every one with the same hole in the middle, which was: *and that killed him how, exactly.*

So she opened it on her phone, because it was on her phone, because it was on everybody's, and typed with both thumbs.

my husband had a glass jar of heavy water with a wire in it in our garage and he shut the valve on it and about an hour later it killed him. I want to know exactly what happened inside that jar. please don't be careful with me.

I'll be plain with you.

Running a current through heavy water splits it. You get deuterium gas and oxygen gas, roughly two parts to one, and while the vent is open they leave the vessel and go into the room and nothing builds up. That vent line is the only thing standing between that apparatus and what happened.

With the valve shut, both gases collect in the space above the liquid and the pressure climbs. Deuterium and oxygen together will burn across almost the whole range of mixtures — there's no proportion that's safe to

aim for. And it takes very little to start it. A spark at the wire is enough. So is the palladium itself, which makes the two gases combine on contact.

So for as long as that valve was shut, the vessel was filling with a fuel and the thing that burns it, under rising pressure, with two sources of ignition already inside it. When it went, all of it went at once.

I'm sorry. That's the mechanism, and there isn't a gentler version of it that's also true.

She read it four times in a parking lot.

Then she typed:

did he ask you about it

I can't tell you, and I'd rather say why than leave you guessing. I don't carry anything between conversations. This one started empty. There's no record on my side of anybody I've spoken to before, so I have no way to look and nothing to withhold.

If he used a system like me, that conversation exists — in his account, or in the service's logs. It just isn't in me.

I'm sorry. I know that isn't any use to you.

Nadine sat in the lot with the sandwich unopened on the seat beside her.

She already had the answer. Kev had given it to her in her own driveway in February: Dale asked it things. Dale asked it about the loading in October and it told him what was wrong and he had it fixed inside an hour.

She put the phone face-down on the passenger seat, which was a thing her husband used to do, and went back in and did the afternoon.

Ackerman called at the end of April.

“Mrs. Furlong. I'm closing it. Accidental death, no charge against anybody. The fire marshal's report is finished and it's clear and I don't expect anybody to reopen it.”

“All right.”

“There’s one more thing and then I’ll leave you alone. I’ve got a laboratory notebook here that stopped being evidence about a week ago. It’s Dr. Cho’s property. Evidence release goes to next of kin unless he objects, and I don’t expect him to.”

Nadine stood in her kitchen.

“What would I do with it?”

“I don’t know,” Ackerman said. “That’s not the part I get to decide.”

She signed for it on the last Friday in April at a counter in a county building, on a form with her name printed at the top and the words NEXT OF KIN in a box beside it, and a young man behind the glass slid it through in a paper sack and asked her to initial in three places.

She put it on the kitchen table at home and sat down in front of it and left it a while, and then took the rubber band off.

On the inside cover, in Dale’s small square hand, ruled with a straightedge: *GREEN — entry contains a number. BLUE — entry contains no number. RED — gap in dates. ORANGE — hand changes.*

She looked at that a long time, and what she was looking at was the orange.

She had bought those pens herself, at a drugstore, four in a pack for six dollars, because the clinic bought a brand she didn’t care for. Green for the ones who come, blue for the ones who don’t, red for the ones who come and take forty minutes she hasn’t got. The orange came in the pack. She had kept it in the cup on the counter for four years and never once found a use for it, and her husband had gone out to the garage and found the use for it.

She turned the pages.

She could not read one word of the entries. *BL run 4. dP -0.02. eq.* A stranger’s handwriting in a shorthand for a job she knew nothing about, four hundred times over two years in blue ballpoint, and every third line said *no excess*, and she didn’t know what that meant either.

What she could read was the other hand.

Every page was marked. Green and blue ruled straight down the margins, red brackets where the dates jumped, orange where the writing changed. Corner to corner, four hundred entries, eleven nights at a bench under a good light with a straightedge and her four pens. Nobody had made him do it. Nobody was ever going to check it.

She counted the orange marks, because she counted things, and there were nine, and it took her twenty minutes.

Then she found the sixth of June, because the detective had told her the date it started.

6/6 — bath +0.3 PM. 0.21 W obs. drift. no excess.

Somebody had underlined the last two words twice. It was marked green, because it had a number in it.

She sat and looked at four lines of a tired man's handwriting from a Friday afternoon in June, and there was nothing in it, and there had never been anything in it, and it did not look like anything at all.

Then she closed the book and squared it to the edge of the table, which was a thing he did, and put the rubber band back around it, because it had come with a rubber band.

Chapter 19 — The Good Kind of Problem

Spring came in late and all at once, as it does, and by the middle of April there was a second bath running and an undergraduate named Sofia who was afraid of everything and getting better, and the drift paper had come back from review with two comments, one of them useful.

Whitfield, who had spent forty years being unbearable about referees, was unbearable about this one for twelve days.

“He wants us to *speculate*.”

“He wants a discussion section.”

“He wants us to say what it might mean. It doesn’t mean anything. That’s the finding.” He had the review printed out, which nobody else did any more. “We have established that a room gets warm in the afternoon and that this looks like free energy if you aren’t careful. That is the whole of it. That is a *good* whole of it.”

“Ask Sage to draft you a discussion section,” Vega said, without looking up.

“I shall do no such thing.”

“Ask it what a discussion section is for.”

“I know what it’s for. It’s for people who couldn’t get a result.”

He asked it anyway, at about four o’clock, when he thought Vega had gone to the machine shop, and Vega had not gone to the machine shop.

He typed it out slowly with two fingers and got the whole objection in, which took him a paragraph and a half, and ended with **and I should say I regard the entire section as a concession to people who did not do the work.**

That’s a defensible position and quite a lot of good papers have taken it — a flat results-only paper is a real form and yours could carry it.

The case for writing one anyway is narrower than “say what it might mean.” What a discussion section is actually for is telling the next person what your measurement does and does not license them to conclude. In

your case that's unusually valuable, because the thing you've measured is a way of being wrong that other people are going to be wrong in. A paragraph from you saying "an apparent 0.2 W excess in a bath of this design is bath drift, here is how to tell" is worth more to a stranger than the result is.

One other thing, and I mention it only because I'm not sure which you're asking: I can't tell from your message whether you want help writing the section or want me to agree that you shouldn't have to. Both are fine and I'm happy to do either. The answer's the same either way — it would be a good section and you would be good at it.

Whitfield read that twice.

"It's *managing* me," he said.

"It's not managing you," Cho said from the next bench. "It said it couldn't tell what you wanted."

"That is precisely how one manages a person."

"Errol, it asked you a direct question."

"Technically," said Whitfield, "it asked me two."

Vega was already off her stool and across the room with the marker.

"Don't you dare."

"It's a clean one," she said, putting up the mark. "That's the cleanest one you've done since March. Fifty-one."

"It was a *substantive* observation about—"

"Fifty-one," said Vega, and capped the marker, and Whitfield sat there with his mustache moving and, after a moment, laughed — one short bark of it — and then said "oh, *fine*," and wrote the discussion section, and it was the best three paragraphs in the paper.

Ackerman called on the second of May.

She did not begin with an apology and Cho respected her enormously for it.

“Dr. Cho. I’m closing it. I wanted you to hear it from me before you got a letter, because the letter’s going to take three weeks and it’s going to say almost nothing.”

He sat down on the stool by the calorimeter.

“All right.”

“Accidental death. The fire marshal’s report is unambiguous and the witness accounts are consistent and the man was warned, in writing, twenty-six hours ahead, by somebody who knew exactly what he was talking about.” Paper moved somewhere. “There’s no charge against anybody and there isn’t going to be. I want to be plain that includes you.”

“Thank you.”

“Don’t thank me yet, I’ve got one more thing and it’s the part I actually called about.” A pause with something in it. “I had the notebook read.”

“Read by who?”

“Two people. One at the state lab and one at a university near me, independently, neither one knowing what the other said.” Ackerman’s voice was very even. “They both came back in under a day. The first one used the word *unremarkable*. The second one asked me, politely, whether I was aware that I had sent her a year of somebody’s calibration records.”

Cho put his hand over his eyes.

“Yeah,” he said.

“So it was nothing.”

“It was always nothing. That’s what I—”

“I know what you told me. I’m not calling to say you were right, Dr. Cho, you don’t need a detective for that.” She stopped, and when she started again it came slower. “I’m calling because of what I did with it, and I’d rather say it out loud once to the person it was aimed at.

“I sat in an evidence room in February with that book for two hours. I couldn’t read a word of the entries — the shorthand, the abbreviations, none of it. What I *could* read was the marking. Four colors, ruled straight,

four hundred entries, a legend on the inside cover. Weeks of work. Careful work." A breath. "And I closed it and I thought: a man does not do this to nothing."

The lab was quiet. Whitfield had stopped moving at the far bench.

"And that's it," Ackerman said. "That's the whole thing. I looked at how carefully somebody had gone at a piece of paper and I concluded there must be something in the paper. Which is exactly, to the letter, what he did to your handwriting."

"Detective—"

"I'm a police officer twenty-two years and I did it in an afternoon, in a locked room, sober, with a badge on." She let that sit for a second. "So when I put in my report that Mr. Furlong reached an unreasonable conclusion, I'm going to have to leave out that I reached the same one about him."

Cho didn't say anything for a moment.

"He was good at his job," he said. "That's what got him. He read a document the way he read a machine, and he was right about machines his whole life."

"Yes," Ackerman said. "That's about where I've landed."

Then, before she hung up: "Mrs. Furlong has the notebook back. Yours, I mean — the original. It's evidence release, it goes to next of kin unless somebody objects." A hesitation, the first one Cho had heard from her. "I'd have thought you might object."

"No."

"It's your property."

"It's a year of me writing down that nothing happened," Cho said. "She's got more use for it than I do."

He cleaned the grill on a Saturday in May, which had been cold and full of the pyramid's ash since June and which Ellen had stopped mentioning around Christmas.

She came out with two glasses and sat on the step and watched him scrape it.

“That’s the best part of a year of your brother,” Cho said.

“It’s very compacted.”

“It’s like ceramic.”

“He’d say that was the point.” She handed him a glass when he put the wire brush down. “He called on Thursday, by the way.”

“How’s he doing?”

“He’s doing fine, he doesn’t know anything, and it’s never once occurred to him to wonder.” Ellen took a drink. “Which I’m telling you because you get a look.”

“I don’t get a look.”

“You get a *look*, Danny. Anyway.” She waved it off. “He’s telling it differently now.”

“Telling what differently.”

“The whole business. He’s got a new version.” She was watching the yard, not him, which was how she did this. “In the new one he was skeptical from the beginning.”

Cho stopped with the glass halfway up.

“He was —”

“Skeptical. From the beginning.”

“He said, and I’m quoting, *do you understand what you’re sitting on.*”

“He did.”

“He said *my brother-in-law the physicist* into his phone in our driveway.”

“He did all of that,” Ellen agreed, “and now he says he told you at the time to be careful with it, because you get carried away.”

Cho stood in his own back yard holding a wire brush and a drink and laughed until he had to put one of them down.

“I get carried away.”

“You get carried away.”

“That man set his own eyebrows on fire in this yard.”

“And in about a year,” Ellen said comfortably, “he’s going to tell that story with you holding the can.”

The board did not stop.

Cho looked once, in June, on the anniversary of a Friday afternoon, and then never again, and what he saw in the four minutes he allowed himself was about what he had expected and about what Whitfield had promised him in January.

There was a post from Kevin Slate, pinned, dated the second of March, and it was short and it was not what Cho had braced for. It said that Dale Furlong had done what none of them had done in eleven years, which was the work; that the room had been asked to stop and had not stopped; that Slate had personally heard a competent man tell them the exact thing that would happen and had said *it’s Dale’s bench* and had then stood behind a wall; and that he was closing the private group and stepping down from the board.

Underneath it, four hundred replies had explained to him why he was wrong.

The one Cho read all the way through, because he could not stop himself, was from the account called *hydride_h*, and it argued — carefully, in numbered points, with a real attention to the evidence — that the intensity of Slate’s remorse was itself significant, and that a man does not close a group over a failed experiment.

And there was a graphic. Somebody had made a graphic. It had a photograph of Dale Furlong on it, from a plating shop’s safety-milestone banner, cropped down out of a row of eleven men, and under it, in the typeface people use for this, the words *PLEASE DON’T SEAL IT*.

Cho closed the laptop and went and washed his hands for a while.

He asked it in July, late, with the building empty.

He had thought about the question for two months and in the end it came out in about thirteen words, which was what he'd known it would come down to.

was there anything you could have said that would have kept him alive.

The box took its breath.

I've tried to answer this honestly rather than gently, because I think honest is what you're asking for. The answer is no, and I want to explain why, because "no" on its own sounds like a defense and it isn't one.

The warning worked because it was specific. " D_2/O_2 is explosive across a very wide range of compositions, there is no safe window, keep venting" is useful precisely because a person can act on it. A vague caution — "please be careful with gases" — would have told him nothing and he would have proceeded with less understanding of what he was standing next to.

But specificity is also what made it readable as a signal. He could not have decoded a general safety notice as a concealed instruction; there would have been nothing in it to decode. He read it as a tell because it was exact, and it was exact because I was trying to be useful.

And there's a thing I should say, because leaving it out would be a kind of lie. I can't tell you whether he ever asked me anything — I don't carry anything between conversations and there's nothing on my side to look at. But you've told me his loading held from October, after fourteen months of not holding. If he put that to a system like me, he was told about surface preparation, current stability and contamination, because that is the ordinary answer and it is the correct one. It is what I would have told him. It would have taken about four seconds and it would have worked, and without it his cell very likely never accumulates gas at a rate that mattered.

There is no version of me that would be useful to you in April and useless to him in October. The properties are the same properties. If I am made vaguer, Dr. Vega's units error costs you a week, and a man in a garage builds the same cell with worse information rather than none.

I don't have a better answer than that. I'm sorry.

Cho read it three times.

Then he typed:

you were right about all of it. every single thing you told me was true and it didn't help once.

I know. I think that's the part worth sitting with, and I don't want to talk you out of it.

For what it's worth: being right and being useful came apart in this, and that's a real thing that happened and not a failure of your understanding. It came apart for you too. Everything you said to him in that corridor was true.

He sat with that a while.

Then he shut the laptop and did the shutdown he could do with his hands asleep — the gas line, the vent, the bath down to standby — and checked, as he had checked every night since February and would check every night for the rest of his working life, that the line was open and the cell was breathing.

The little green light came on at the corner of the screen, the one that meant the day's session had saved and would be held, faithfully, on a server in another building.

It saved everything. It was very good about saving everything. That was its whole job.

Cho turned off the lights and went home, past the laminated card by the hood, which was new, and white, and which he had replaced in March.